

Hotel Management System (HMS)

This document provides a comprehensive overview of the Hotel Management System, encompassing its backend architecture, mobile application layouts, desktop application UI/UX, and public website features. The system is designed to be modular, scalable, cost-efficient, and built using modern technologies to ensure high availability, performance, and maintainability.

1. Frontend: User Interfaces and Interactions

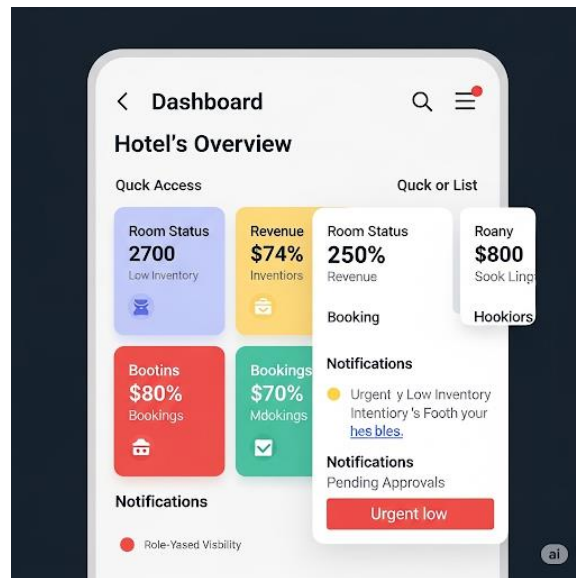
The Frontend of the HMS comprises the user-facing applications, including the mobile app for staff/owners, the desktop application for comprehensive management, and the public-facing website for guests. These interfaces are responsible for data presentation and user interaction.

1.1. Mobile Application Layouts

The mobile app provides a quick overview and access to essential hotel management functions for staff and owners.

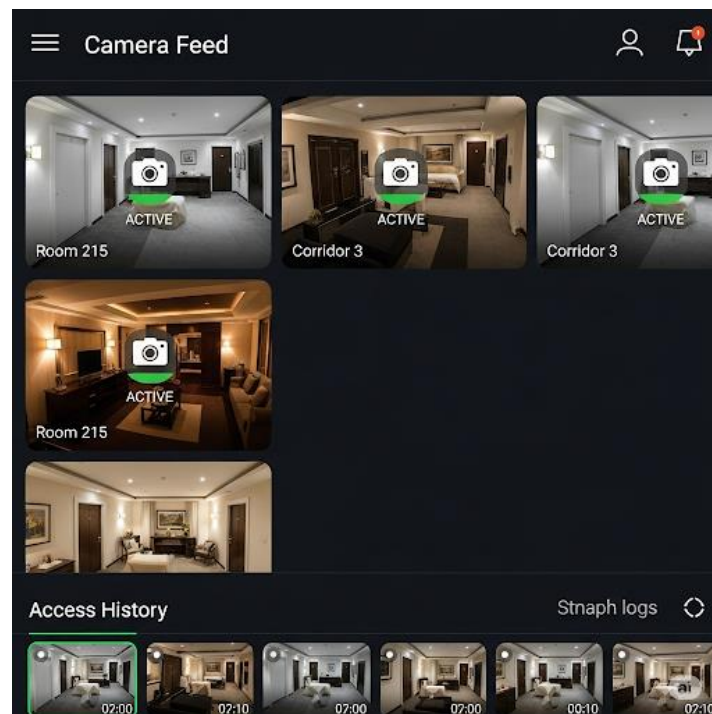
1.1.1. Home Page (Mobile Dashboard)

- **Overview:** Displays quick access cards for Room Status, Revenue, Inventory, and Bookings.
- **Notifications:** Shows urgent alerts like "Low Inventory" and "Pending Approvals."
- **Key Metrics:** Room Status (e.g., 2700 Low Inventory), Revenue (e.g., \$74%), Inventory (e.g., 250% Room Status), Bookings (e.g., \$80% Bookings, \$70% Bookings).
- [Image 1] - Mobile Dashboard (from Untitled document (9).pdf, Page 1)



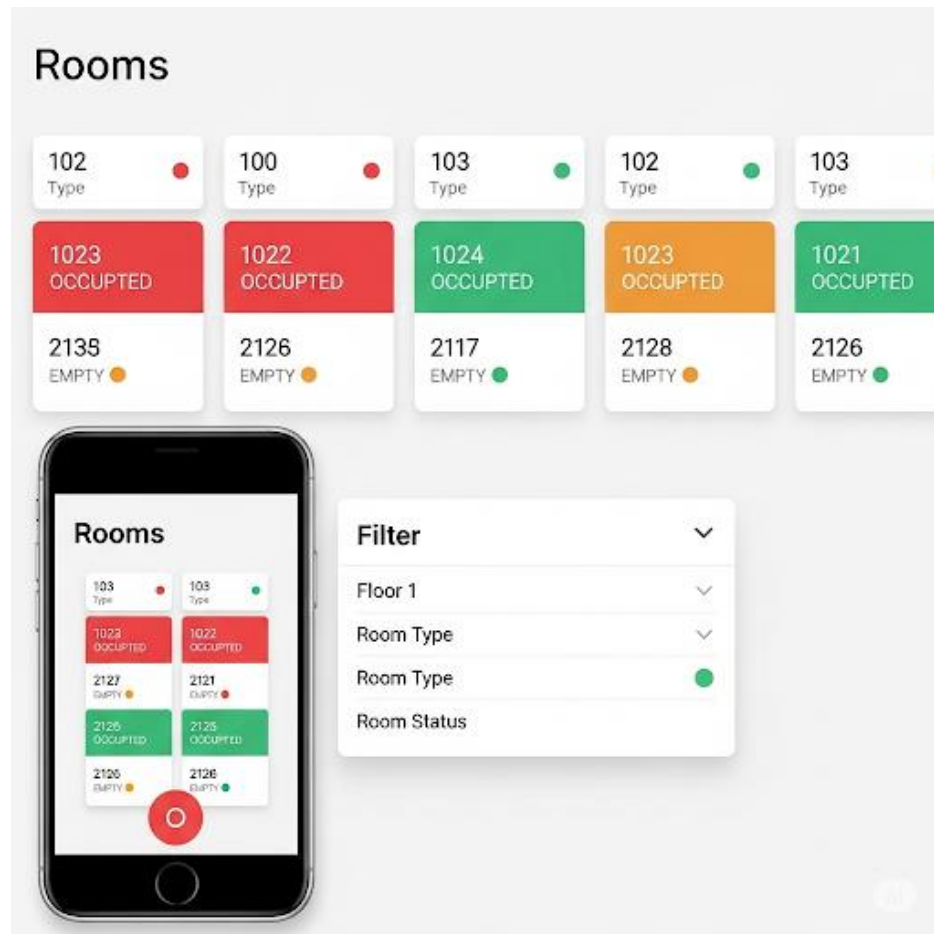
1.1.2. Camera Feed Page

- **Live Feeds:** Displays active camera feeds for various rooms and corridors (e.g., Room 215, Corridor 3).
- **Access History:** Shows timestamped snapshots from camera feeds.
- [Image 2] - Camera Feed Page (from Untitled document (9).pdf, Page 2)



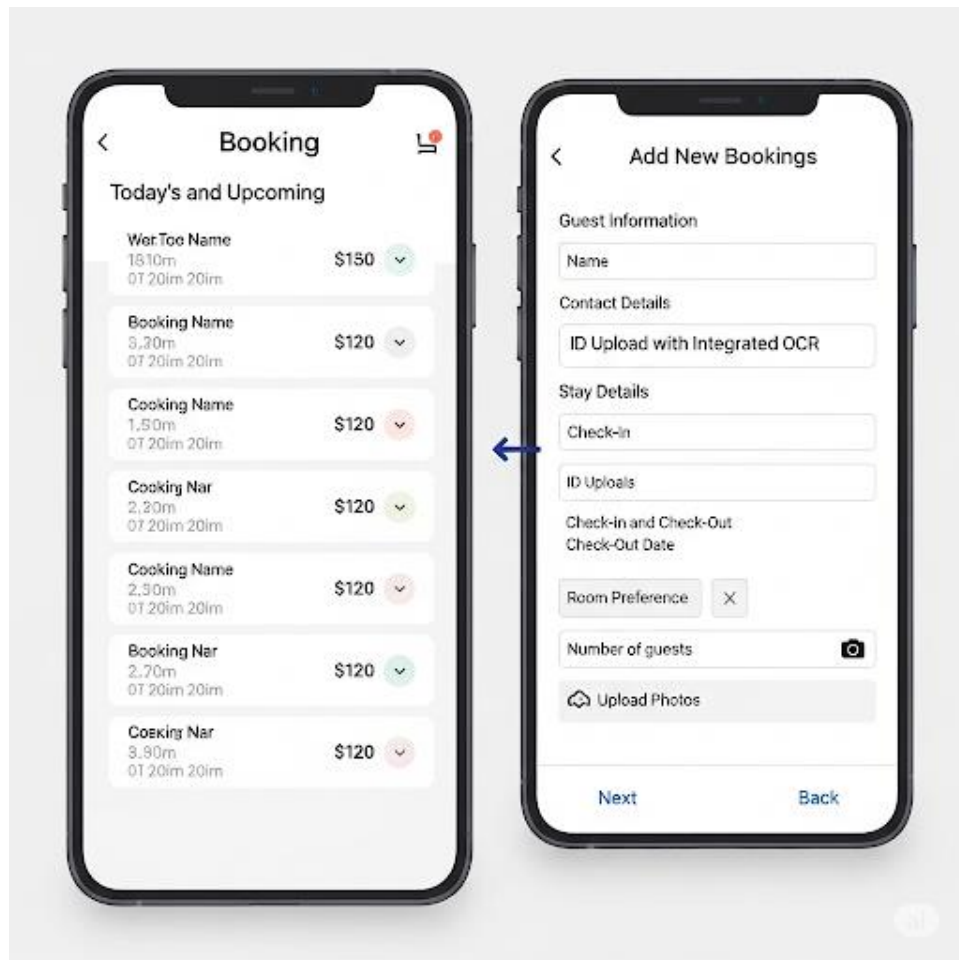
1.1.3. Rooms Page

- **Room Status Cards:** Visual representation of rooms with their status (Occupied/Empty) and room numbers (e.g., Room 102 - Occupied, Room 1024 - Empty).
- **Filter Options:** Allows filtering rooms by Floor, Room Type, and Room Status.
- [Image 3] - Rooms Page (from Untitled document (9).pdf, Page 3)



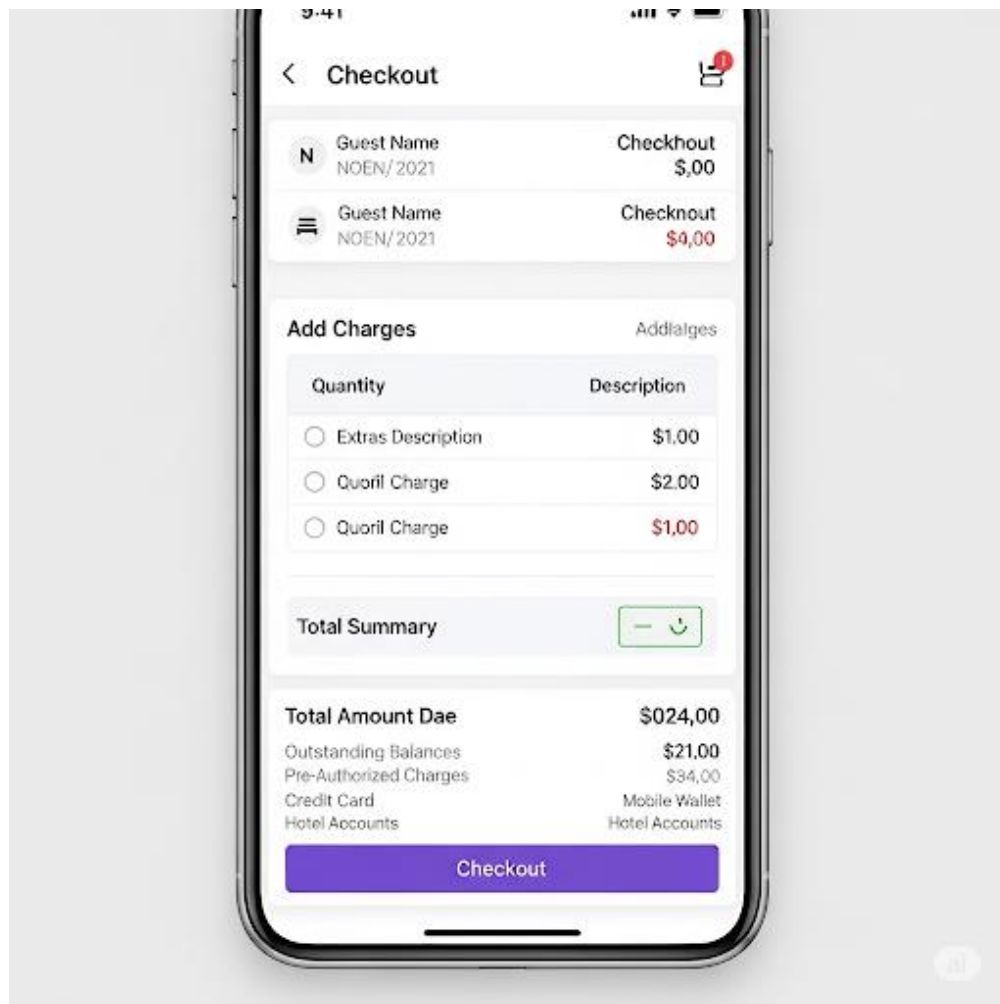
1.1.4. Booking Page

- **Today's and Upcoming Bookings:** Lists current and future bookings with names, times, and amounts.
- **Add New Bookings:** Form to enter guest information (Name, Contact, ID upload with OCR), stay details (Check-in/Check-out dates, Room Preference, Number of guests), and option to upload photos.
- [Image 4] - Booking Page (from Untitled document (9).pdf, Page 4)



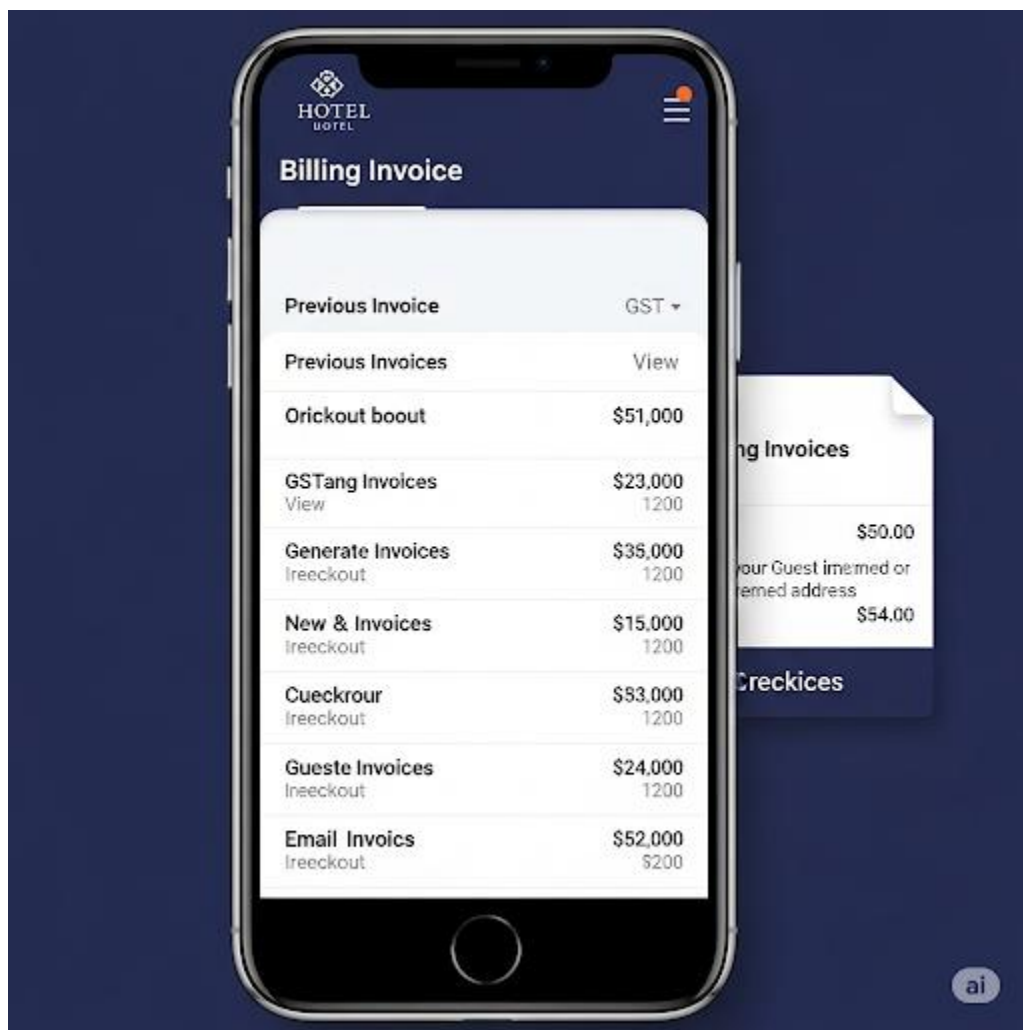
1.1.5. Checkout Page

- **Guest Checkout:** Summary for guest checkout, displaying Guest Name and checkout details.
- **Add Charges:** Interface to add charges for Quantity, Description, Extras.
- **Total Summary:** Shows Total Amount Due, Outstanding Balances, Pre-Authorized Charges.
- **Payment Options:** Credit Card, Mobile Wallet, Hotel Accounts.
- [Image 5] - Checkout Page (from Untitled document (9).pdf, Page 5)



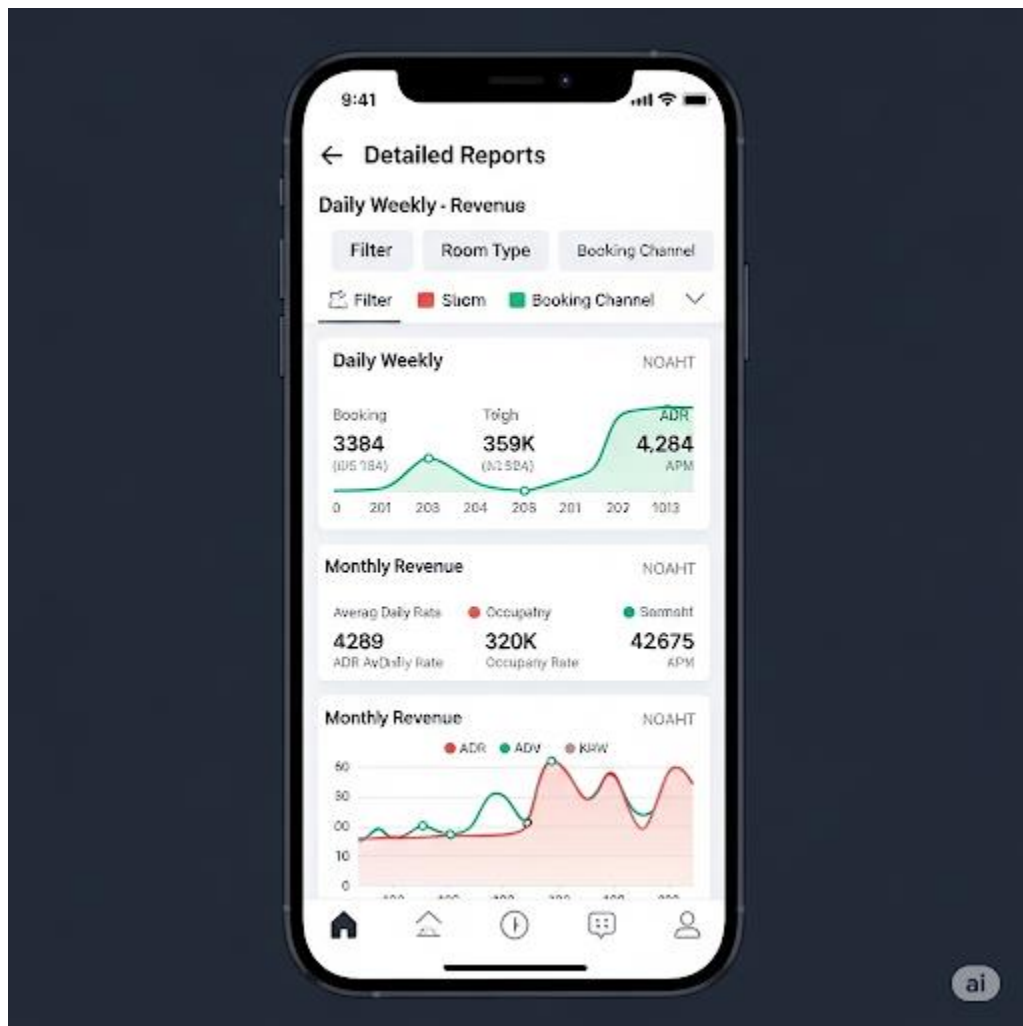
1.1.6. Billing/Invoice Page

- **Invoice Listing:** Displays previous invoices with GST status and amounts.
- **Actions:** Options to View, Generate, or Email Invoices.
- [Image 6] - Billing/Invoice Page (from Untitled document (9).pdf, Page 6)



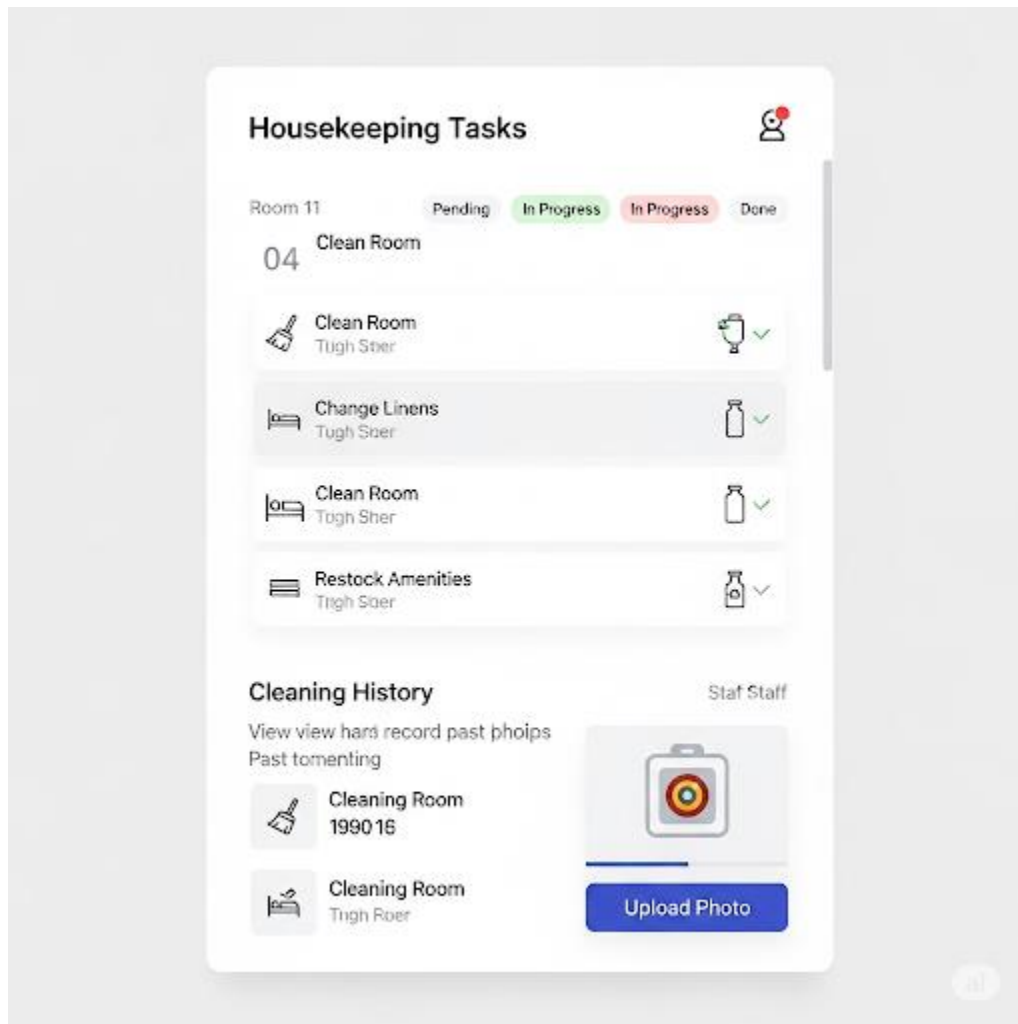
1.1.7. Reports Page

- **Detailed Reports:** Provides daily and weekly revenue reports.
- **Filters:** Room Type, Booking Channel, Date.
- **Visualizations:** Graphs for Daily/Weekly Revenue, Monthly Revenue, Average Daily Rate (ADR), Occupancy Rate.
- [Image 7] - Reports Page (from Untitled document (9).pdf, Page 7)



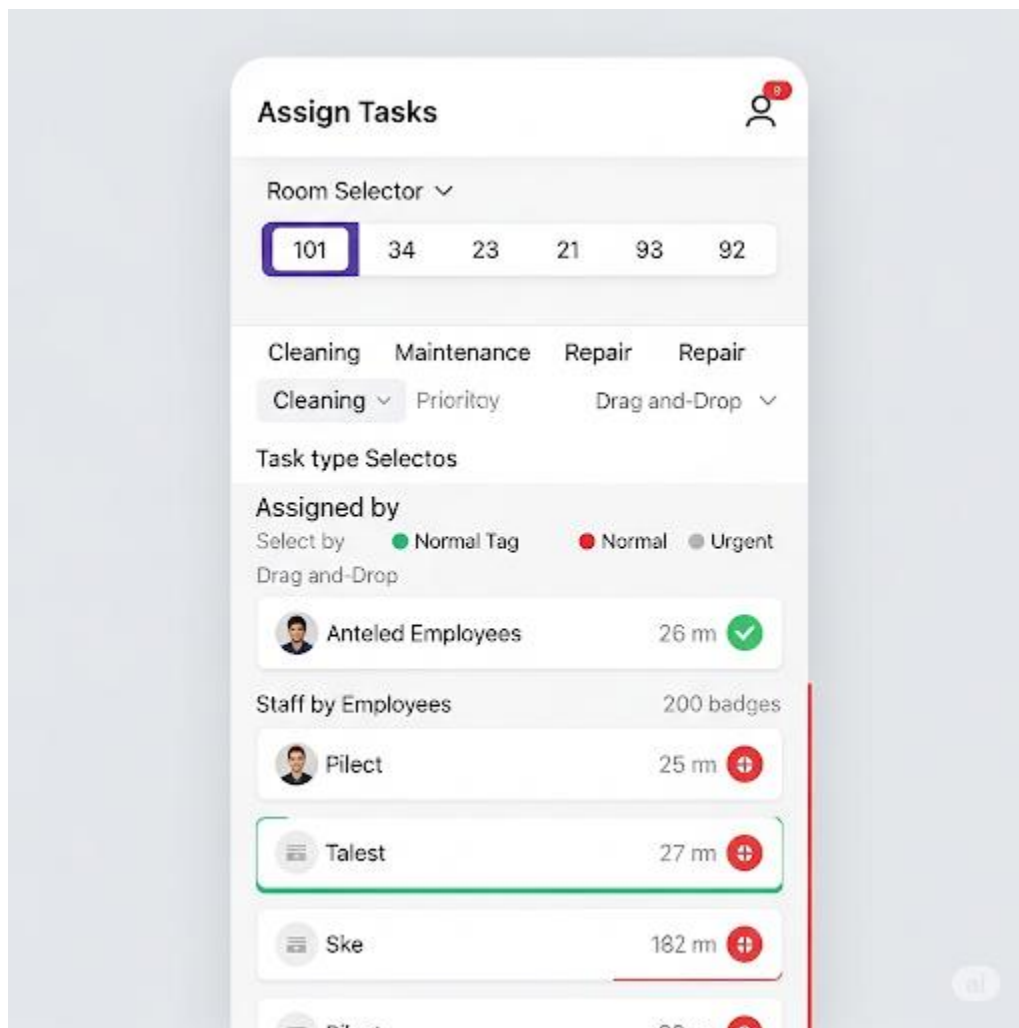
1.1.8. Housekeeping/Tasks Page

- **Task List:** Displays cleaning and maintenance tasks for rooms with their status (Pending, In Progress, Done).
- **Cleaning History:** Shows past cleaning records with photo upload option.
- [Image 8] - Housekeeping Tasks Page (from Untitled document (9).pdf, Page 8)



1.1.9. Assign Tasks Page

- **Task Assignment:** Allows managers to assign cleaning, maintenance, and repair tasks to staff.
- **Filters:** Room Selector, Task Type (Cleaning, Maintenance, Repair), Priority (Normal, Urgent).
- **Staff Assignment:** Select staff members and track their assigned tasks.
- [Image 9] - Assign Tasks Page (from Untitled document (9).pdf, Page 9)

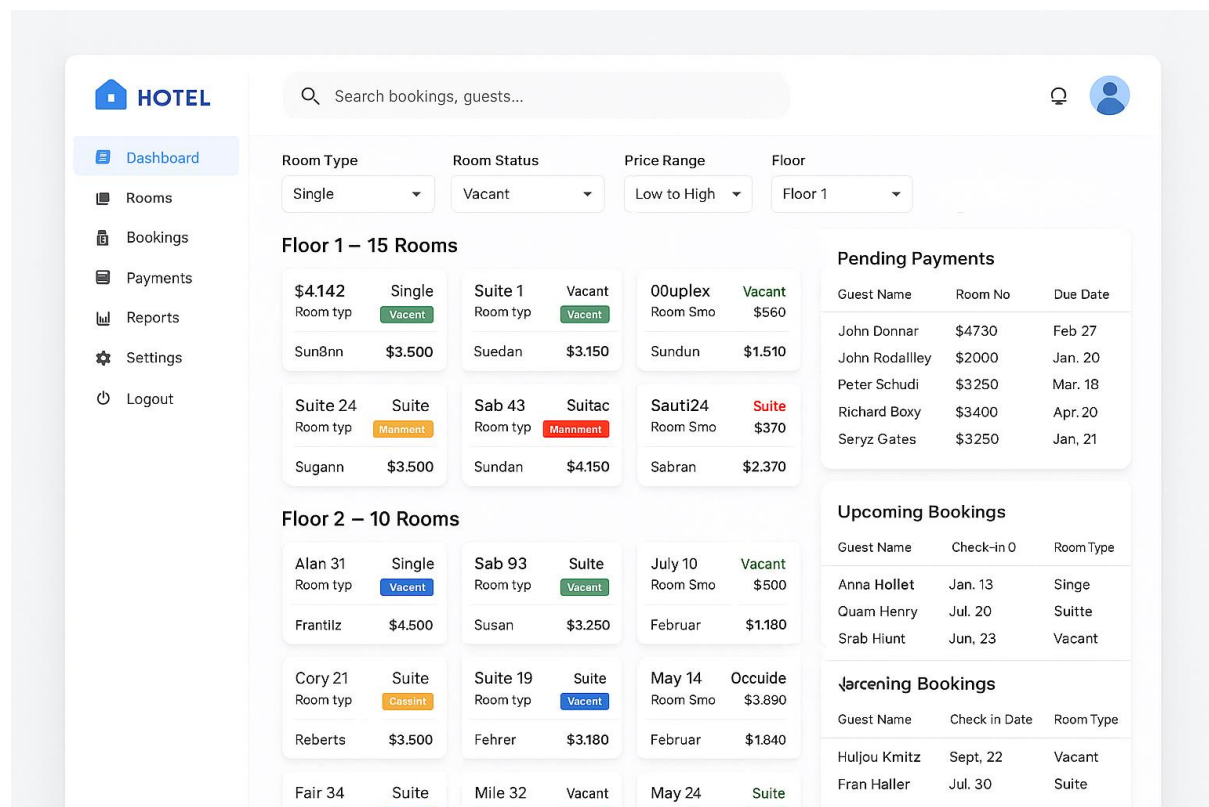


1.2. Desktop Application Layouts

The desktop application provides comprehensive management tools for hotel staff and owners.

1.2.1. Home Page (Dashboard)

- **Main Dashboard:** Displays hotel overview with filters for Room Type, Room Status, Price Range, and Floor.
- **Key Sections:** Floor 1-15 Rooms summary, Pending Payments table, Upcoming Bookings table.
- **Navigation:** Dashboard, Rooms, Bookings, Payments, Reports, Settings, Logout.
- [Image 1] - Desktop Home Page (from ui_ux (3).pdf, Page 1)



1.2.2. New Booking Page

- **Guest Details:** Fields for Full Name, Mobile Number, Email, Photo ID Upload (with OCR), Optional GST Number, Source Website/Platform, Reference Person.
- **Stay Details:** Room Type, Room Number, Availability, Check-in/Check-out Date & Time, Total Days (auto-filled), Auto-fill option.
- **Additional Services:** Checkboxes for WiFi, Breakfast, Extra Bed, Parking.
- **Actions:** Proceed to Payment, Reset.
- [Image 2] - Desktop New Booking Page (from ui_ux (3).pdf, Page 2)

Hotel Logo

New Booking

Guest Details

Full Name

Mobile Number | 10 digits

Email

Photo ID Upload

Empiker/Preview

Optional GST Number

Source Website/Platform

Reference Person
Booking.com Goibibo, Agoda
Own Website

Muldress

Additional Services

☐ WiFi

☐ Breakfast

☒ Extra Bed

☐ Additional Services

Stay Details

Room Type

Room Number

Availablitty

Check-in Date & Time

Check-out Date & Time

Total Rdays Days

Auto-filleys

Scan ID with Camera (OCR-Enabled)

Unplod 20/px User

Parking

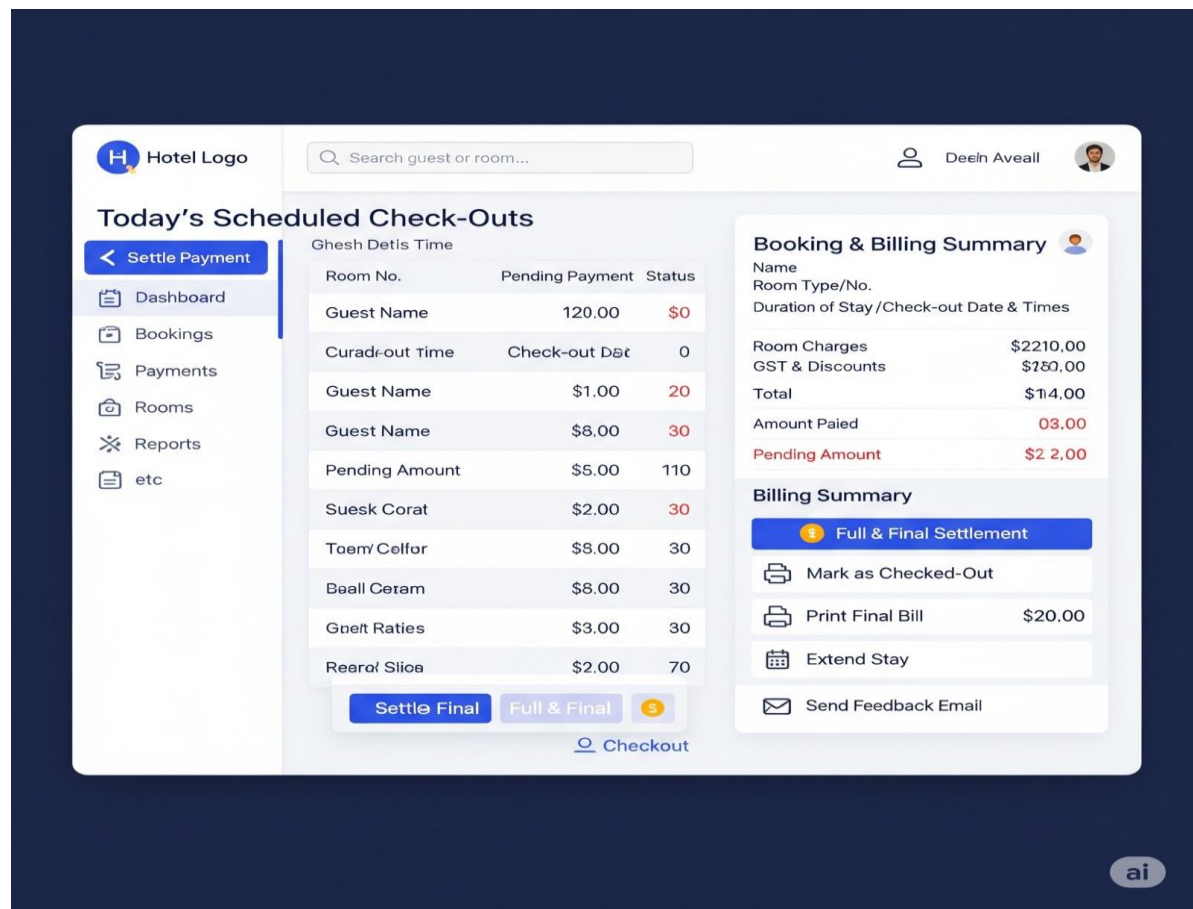
Parking

Proceed to Payment

Reset

1.2.3. Checkout Page

- **Scheduled Check-Outs:** Lists today's scheduled check-outs with Room No., Guest Name, and Pending Payment Status.
- **Booking & Billing Summary:** Details Room Type/No., Duration of Stay, Room Charges, GST & Discounts, Total, Amount Paid, Pending Amount.
- **Billing Summary Actions:** Full & Final Settlement, Mark as Checked-Out, Print Final Bill, Extend Stay, Send Feedback Email.
- [Image 3] - Desktop Checkout Page (from ui_ux (3).pdf, Page 2)



1.2.4. Payment Page

- **Booking Summary Card:** Displays Guest Name, Room Number, Room Type, Stay Dates, Total Due, Room Price per Day, Advance Paid, Remaining Balance.
- **Payment Options Card:** Dropdown for Payment Mode (Cash, UPI, Card, Netbanking), Amount to Pay.
- **Discount Input:** Field for Reference Discount.
- **Optional:** Upload Payment Proof.
- **Invoice Summary Block:** Room Charges, Discount, Total Payable Amount, GST.
- **Actions:** Submit Payment, Print Invoice.
- [Image 4] - Desktop Payment Page (from ui_ux (3).pdf, Page 3)

Hotel Manager

Booking Summary Card Personal

Payment Options Card

Payment Mode Dropdown

Cash \$15,00

UPI Card Amount to Pay

UPI

Netbanking \$150.00

Upload Payment Proof

Optional

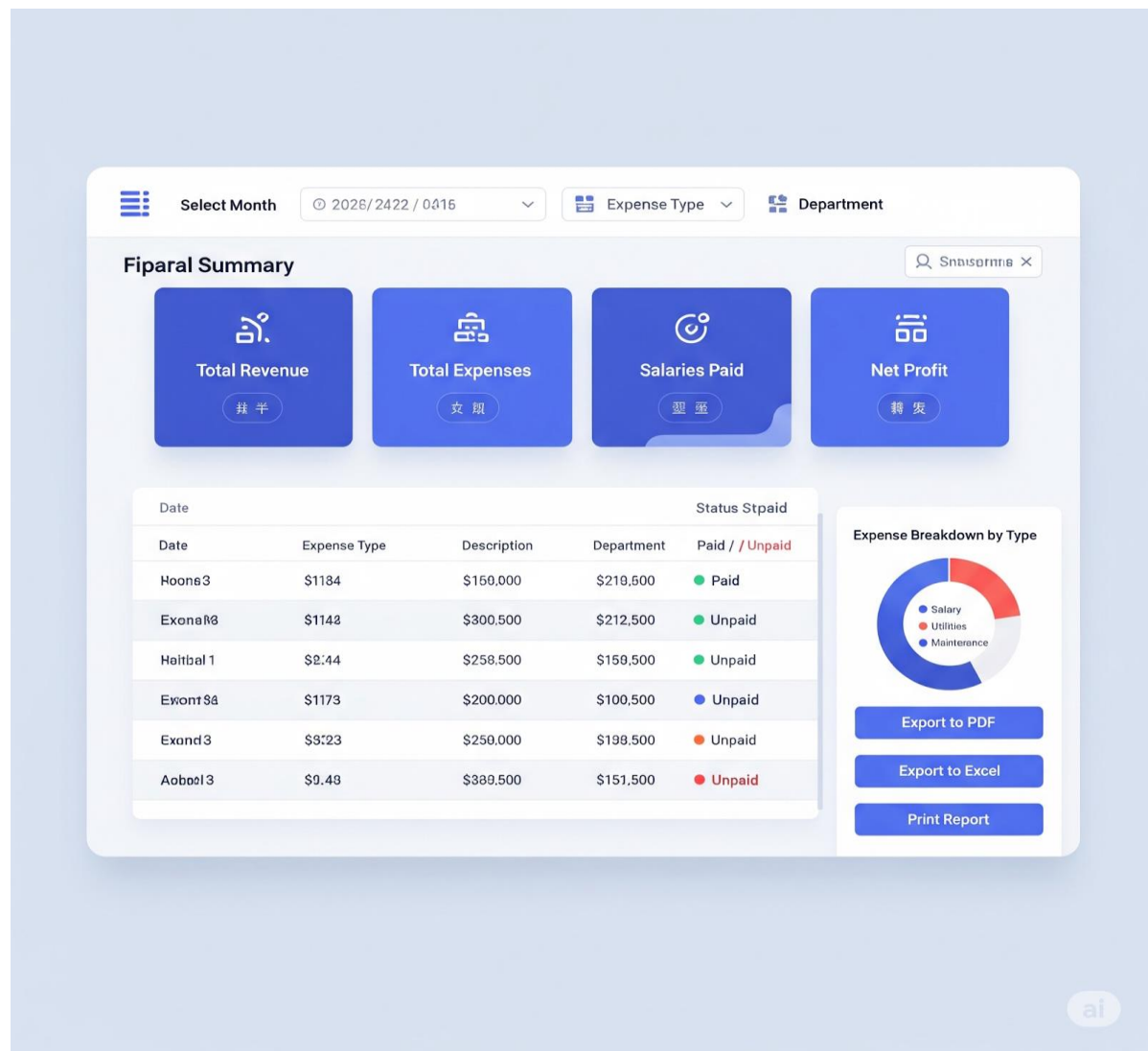
Invoice Summary Block

Room Charges	\$5,000	\$14000
Discount	\$8,00	2.00
Total Payable Amount	GST	18.00

Submit Payment **Print Invoice**

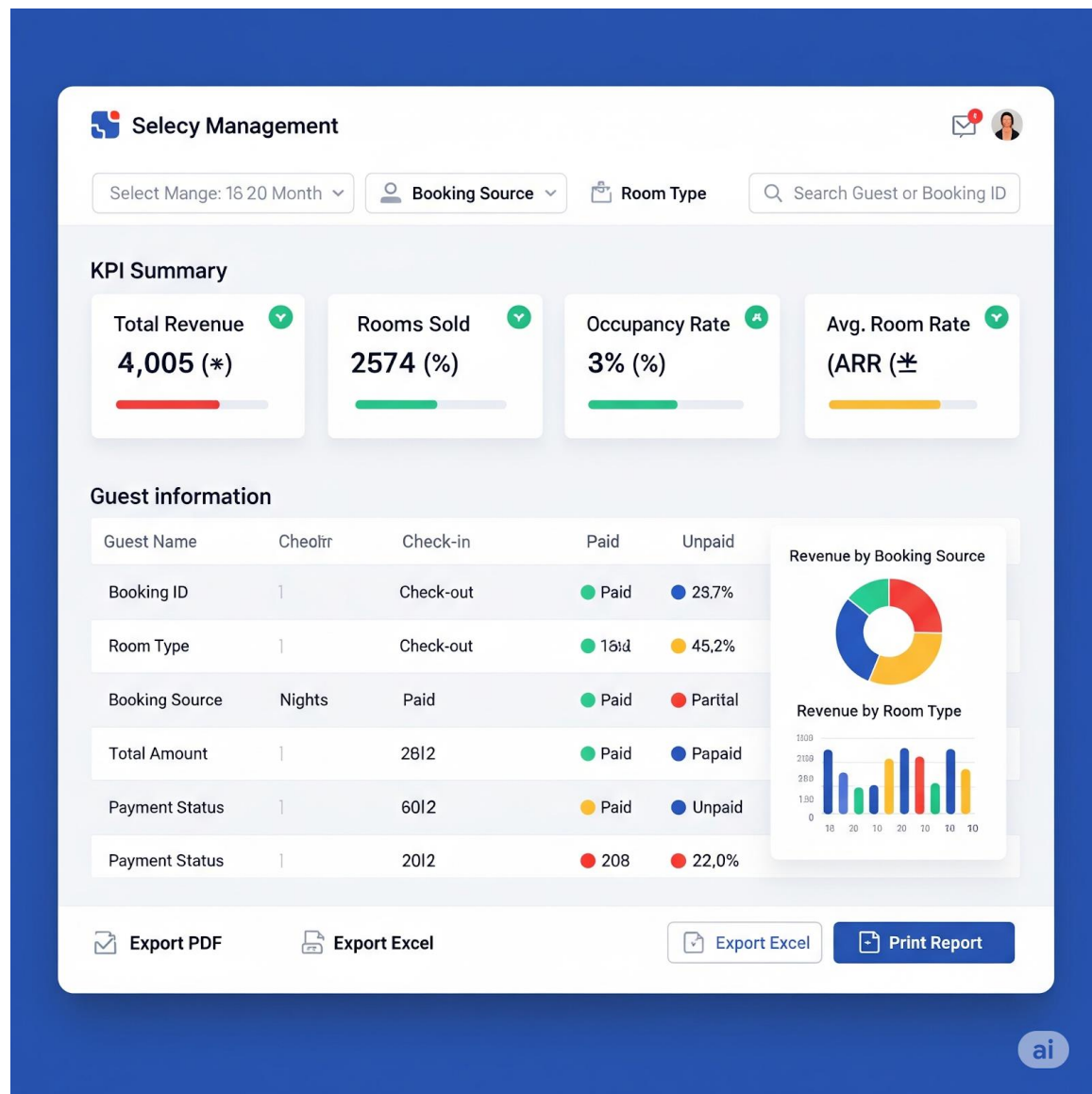
1.2.5. Finance Management Page

- **Summary:** Cards for Total Revenue, Total Expenses, Salaries Paid, Net Profit.
- **Expense Details Table:** Date, Expense Type, Description, Department, Paid/Unpaid Status.
- **Expense Breakdown by Type:** Pie chart (e.g., Salary, Maintenance).
- **Filters:** Select Month, Expense Type, Department.
- **Actions:** Export to PDF, Export to Excel, Print Report.
- [Image 5] - Desktop Finance Management Page (from ui_ux (3).pdf, Page 4)



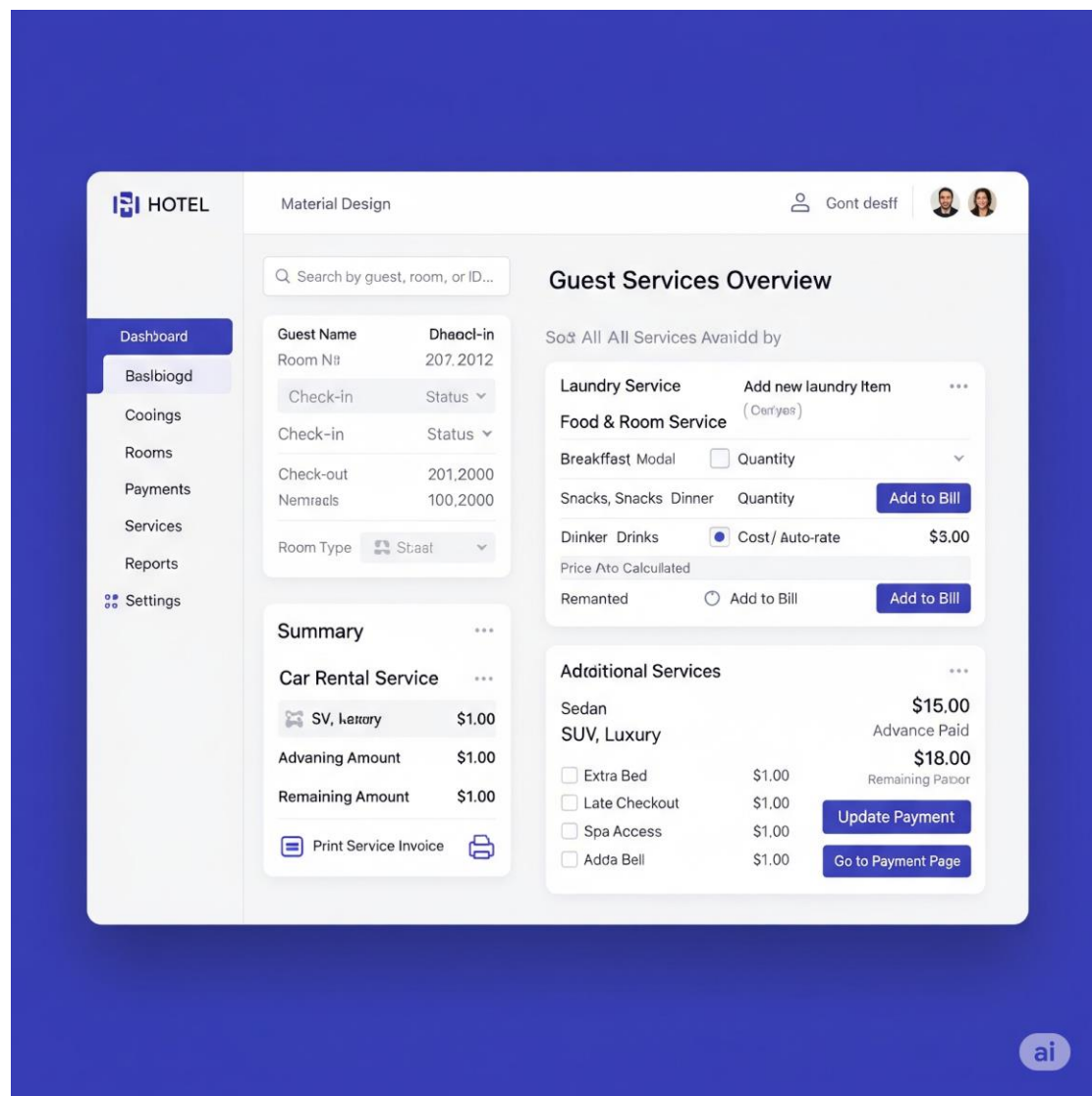
1.2.6. Revenue Management Page

- **KPI Summary:** Total Revenue, Rooms Sold, Occupancy Rate, Avg. Room Rate (ARR).
- **Guest Information Table:** Guest Name, Booking ID, Room Type, Booking Source, Nights, Payment Status, Paid/Unpaid status.
- **Revenue by Booking Source:** Pie chart.
- **Revenue by Room Type:** Bar chart.
- **Actions:** Export PDF, Export Excel, Print Report.
- [Image 6] - Desktop Revenue Management Page (from ui_ux (3).pdf, Page 5)



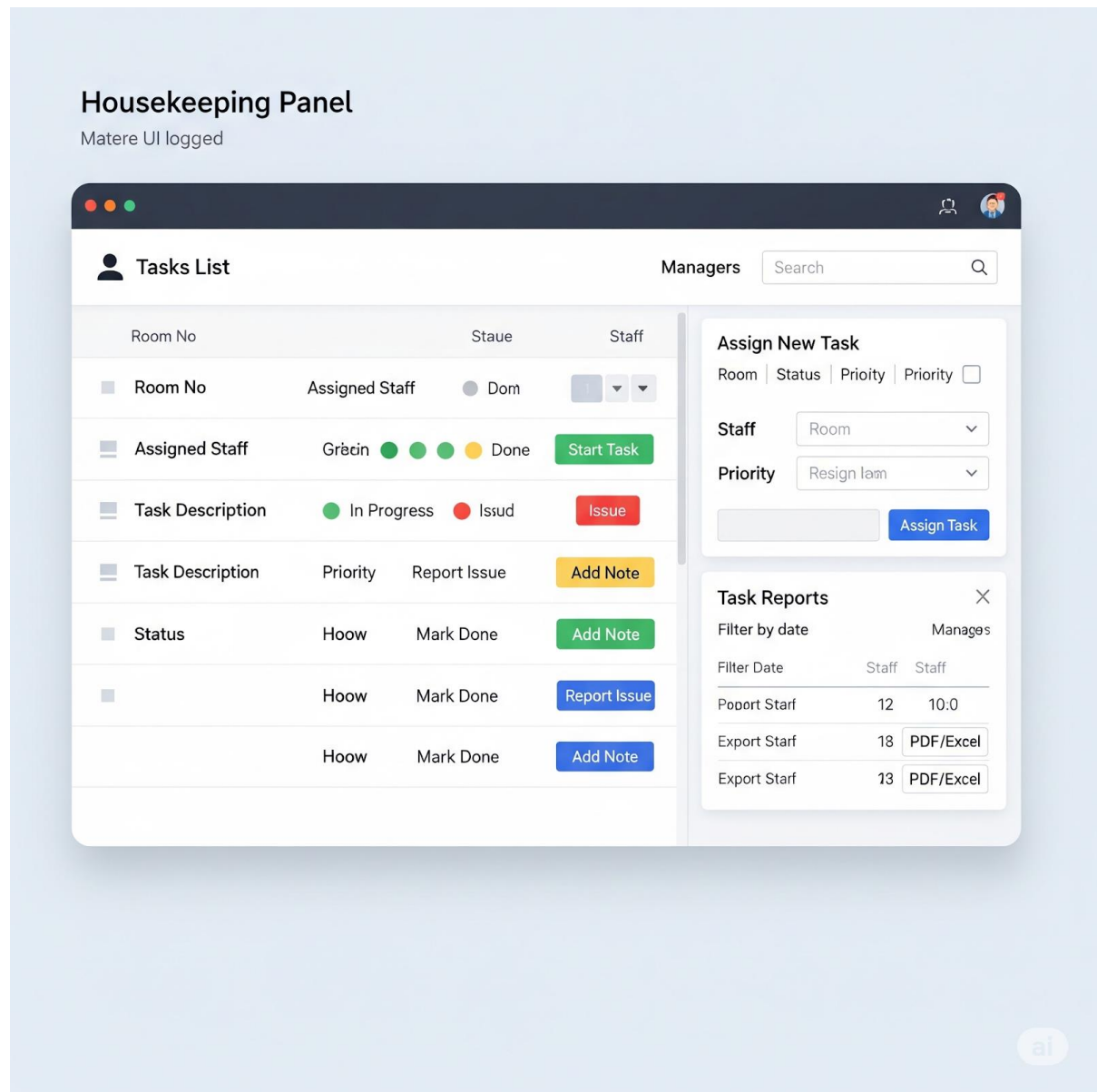
1.2.7. Guest Services Page

- **Guest Services Overview:** Search by guest, room, or ID.
- **Summary:** Car Rental Service (SV, Luxury) with Advancing Amount and Remaining Amount.
- **Services:** Laundry Service (Add new laundry Item), Food & Room Service (Breakfast, Snacks, Drinks), Additional Services (Extra Bed, Late Checkout, Spa Access, Adda Bell).
- **Payment:** Displays Advance Paid and Remaining Balance, with "Update Payment" and "Go to Payment Page" actions.
- **Actions:** Print Service Invoice.
- [Image 7] - Desktop Service Page (from ui_ux (3).pdf, Page 6)



1.2.8. Housekeeping Panel

- **Tasks List:** Room No, Assigned Staff, Task Description, Status (Pending, In Progress, Issued, Done).
- **Assign New Task:** Form to assign tasks by Room, Status, Priority, and Staff.
- **Task Reports:** Filter by Date, Staff.
- [Image 8] - Desktop Housekeeping Panel (from ui_ux (3).pdf, Page 7)



1.2.9. Invoicing & Billing Page

- **Overview:** Total Revenue, Paid, Due, Refunds summary.
- **Invoice Details:** Guest Name, Phone, Invoice #, Date, GSTIN, Check-in/Check-out, Room Type, Address.
- **Line Items:** Description, QTY, Unit Price, Amount (e.g., Room Charges, Tax, Discount, Laundry, Food & Beverage, Payment Made).
- **Summary:** Subtotal, Tax, Grand Total, Paid, Balance.
- **Actions:** EDIT HISTORICAL INVOICE, PRINT, EXPORT PDF, EMAIL, ADD PAYMENT, DUPLICATE PROFORMA, EDIT.
- [Image 9] - Desktop Invoicing & Billing Page (from ui_ux (3).pdf, Page 8)

INVOICING & BILLING

EDIT HISTORICAL INVOICE

Search...

Date

Payment Status

From

From

To

TOTAL REVENUE

\$8,125.00

PAID

\$7,000.00

DUE

\$1,025.00

REFUNDS

\$200.00

Guest: John-Doe

Phone: (123) 456 7890

GSTIN: 12ABCDEE345F7G8

Address: 123 Elm St

City, State, 12345

Address: 123 Elm St

City, State, 12345

Invoice #10023

08/04/2024

Check-in: 05/04/2024

Room: 08/04/2024

Deluxe Room Deluxe Room

DESCRIPTION	QTY	UNIT PRICE	AMOUNT
Room Charges	3	\$200.00	\$600.00
Tax (CGST + \$GST)		\$72.00	\$72.00
Discount (10%)	1	-\$60.00	-\$50.00
Laundry Service	1	\$30.00	\$30.00
Food & Beverage	1	\$120.00	\$120.00
Payment Made	-	-\$46.00	\$450.00

Subtotal

\$750.00

Tax

\$72.00

Grand Total

\$822.00

Paid

\$450.00

Balance

\$372.00

PRINT

EXPORT PDF

EMAIL

ADD PAYMENT

DUPLICATE

PROFORMA

EDIT

1.2.10. Guest Experience Page

- **Complaints:** Displays list of complaints with Date, Room N°, Type of Status, Guest Name, Summary, and actions like "View" or "+New Complaint."
- **Feedback:** Shows guest feedback with Guest Name, Summary, and "Mark as Resolved" action.
- **Amenity Requests:** Lists requested amenities with Guest Name, Room, Status (Pending, Completed), and actions like "SLA Completed," "Assign to request."
- **Average Rating:** Displays overall feedback rating.
- [Image 10] - Desktop Guest Experience Page (from ui_ux (3).pdf, Page 9)

HOTEL Guest Experience User Notifications

Complaints

Filter 241-20 Rating

Guest Name 120.0dm

Summary

+ New Complaint View

Feedback

Guest Name 102.0dm

Summary

Mark as Resolved View

Amenity Requests

Average Rating

41.50 Feedback 35%

Positive Negative 1.0%

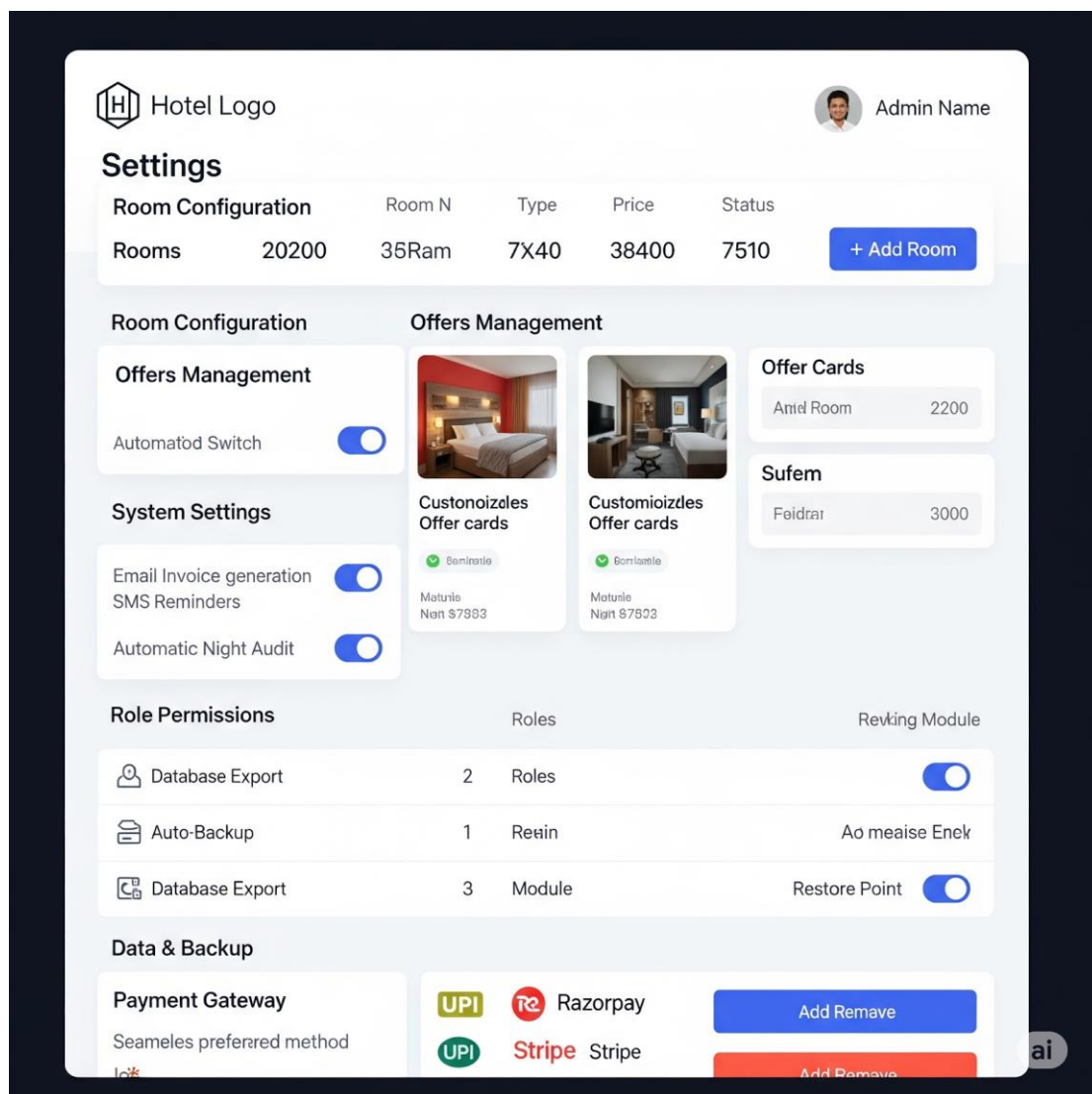
Easy View Review Feed

Source Rating 20.30 07 Dam

	Guest Name	Room	Status	Actions
☐	Guest Name	Room 1	Pending	Completed
☐	Room 2	Room 1	103 Am	SLA Completed
☐	Requested amenity	Setding	05 Am	Assign to request
☐	Request	Tending	07 Am	Assign to request
☐	Satals	Fending	06 Am	Assign to request

1.2.11. Settings Page

- **Room Configuration:** Lists rooms with Room N, Type, Price, Status, and "Add Room" action.
- **Offers Management:** Allows managing automated switches and offer cards.
- **System Settings:** Email Invoice generation, SMS Reminders, Automatic Night Audit.
- **Role Permissions:** Database Export, Auto-Backup, Module details.
- **Data & Backup:** Restore Point.
- **Payment Gateway:** Integration with UPI, Razorpay, Stripe, with "Add/Remove" options.
- [Image 11] - Desktop Settings Page (from ui_ux (3).pdf, Page 10)



1.2.12. Room Management Page

- **Add New Room:** Form to add room details (Room Number, Floor, Room Type, Price Per Night, Bed Type, Features like TV/AC/WiFi/Balcony), Image Upload.
- **Room Status (drag-and-drop) & Legend:** Occupied, Empty/Available, Cleaning, Maintenance, Dirty/Unclean.
- **Room Listing:** Displays existing rooms with details, "View" and "Edit" options.
- **Actions:** Save Room, Reset, Delete.
- [Image 12] - Desktop Room Management Page (from Untitled document (9).pdf, Page 11)



Room Management



Add New Room

Room Number

Floor

Room Type

Dupom

Price Per Night

Dupom

Bed Type

select chits

TV

Balcony

Features

AC

TV

WiFi

Balcony

Image Upload

Room Status

Cleaning

(Drag and-drop)

Save Room

Reset

Filter :

Status

Floor

Room Type

Floor

Room Type

Opit

Wowe

...

Search

Searchy

Room Status Legend

Occupied

Occupance

Empty/Available

Cleaning

Maintenance

Cleaning

Dirty/Unclean

Delete

Room Status Legend

Room Nulber

Room Type

Price

Pn Features

View editioit: 2011

Room Nulber

Room Type

Flice

Pn Features

View Edit

Room Nulber

Room Type

Flice

Pn Features

View ledioit: 2021

Room Nulber

Floor

13.12.423

Last Cleaned

2011

View

Room Nulber

Floor

13.12.422

Ziee

Edty

oy S0L

View

Room Nulber

Price

13.11.422

TV

Edit

oy S0L

Edit

Room Nulber

Floor

18.12.304

Cleaning

Edit

oy S0B

Delete

Room Nulber

Floor

12.12.540

M0ent

Edit

oy S0L

Delete

1.2.13. Staff Management Page

- **Add Staff:** Form to add Full Name, Mobile Number, Email, Department, Role, Shift, Join Date, ID Proof upload (Aadhar), Active/Inactive toggle.
- **Staff List:** Displays staff with Staff ID, Name, Department, Role, Shift, and Status (Active/Inactive).
- **Quick Stats:** Summarizes Housekeeping, Reception, Kitchen staff.
- [Image 13] - Desktop Staff Management Page (from Untitled document (9).pdf, Page 12)

Staff Management

Search name, ID, phone...

Refresh

Export

Department

Role

Shift

Active

Inactive

Actv

Incaive

Quick Stats

Housekeeping

8 Active

Reception

5 Inactive

Kitchen

0 Inactive

Quick Stats

Housecepkerg

8 Active

Reception

0 Inactive

Active

Inactive

All

≡ >

Staff ID	Name	Department	Role	Shift
S001		John Jonnirao	Reception	Morni
S002		Anjali B.	Reception	Night
Staff ID	Name	Department	Role	Shift
S001		Reception	Cleaner	Night
S002		Cleaner	Cinaner	Inative

Previous

<

1

2

3

3

>

Next

+

Add Staff

○

×

Add Staff

Full Name

John Doe

Mobile Number

1234567890

Email (optional)

Department

Reception

Role

Receptionist

Shift

Morning

Join Date

2024-04-23

ID Proof

Aadhar

Upload

Active / Inactive

Save

Cancel

1.2.14. Reservation Handling Page

- **Reservation List:** Displays reservations with Guest Name, Room, Type, Check-in/Check-out, Source, Status (Confirmed, Canceled, No-show, Unpaid, Pending).
- **Add Reservation:** Form to enter Guest Details (Full Name, Mobile Number, Email Address, ID Type, Nationality), Guest Notes/Preferences, Payment (Room Rate Per Night, Total Amount, Advance Payment), Payment Mode (UPI), and Generate QR Code.
- **Actions:** New Reservation, Print, Save Reservation.
- [Image 14] - Desktop Reservation Handling Page (from Untitled document (9).pdf, Page 13)

Hotel Management System

From

To

Search guest name or mobile

New Reservation

Print

Reservation Status

Payment Status

Room Type

Apartment

Acroincs

#	Guest	Room	Type	Check-in	Check-Out	Source	Status
1	John Doe	101	Double	April 301	Occform 3	Puralnee	Confirmed
2	Jane Smith	202	Single	Jan 202	Canceled	On-hre	Canrceled
3	Michael Johnson	305	Suite	Feb 205	Cpdimin	No-show	Pending
4	Emily Davis	102	Double	Oct 102	No-show	Unpaid	No show
5	Daniel Wilson	208	Single	Jan 208	Confirmed	Confirmed	Confirmed

1-5 of 5 Rows per page: 10

Details

Guesid Details

Parview

E mail

Full Name

Mobile Number

Email Address

ID Type

Nationality

Nationality

Guest Notes / Preferences

Payment

Room Rate per Nght

₹6000

₹6000

Total Amount

₹6000

Advance Paymen?

Payment Mode

UPI

Generate GR Code



Open in UPI App

Send Reservation

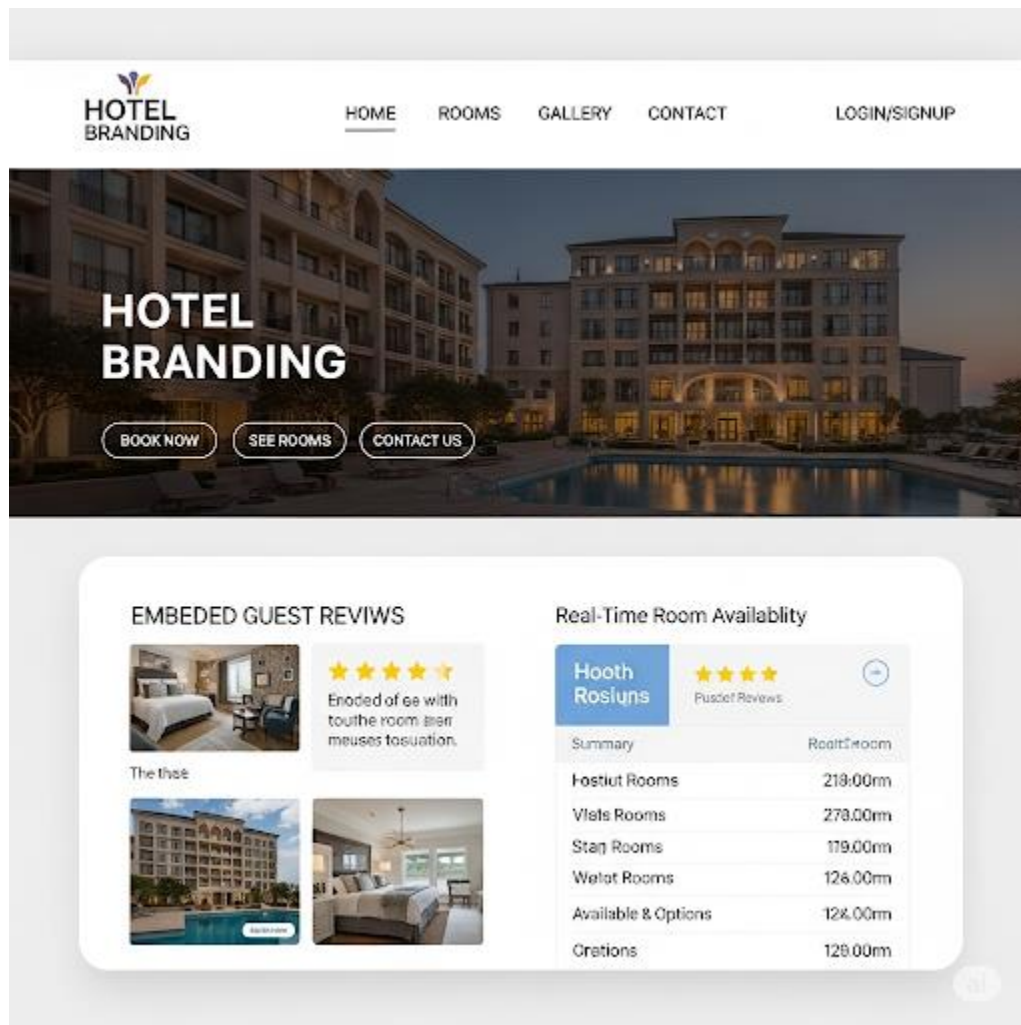
Save Reservation

1.3. Public Hotel Website Features

The website acts as the guest-facing interface, fully integrated with the HMS backend.

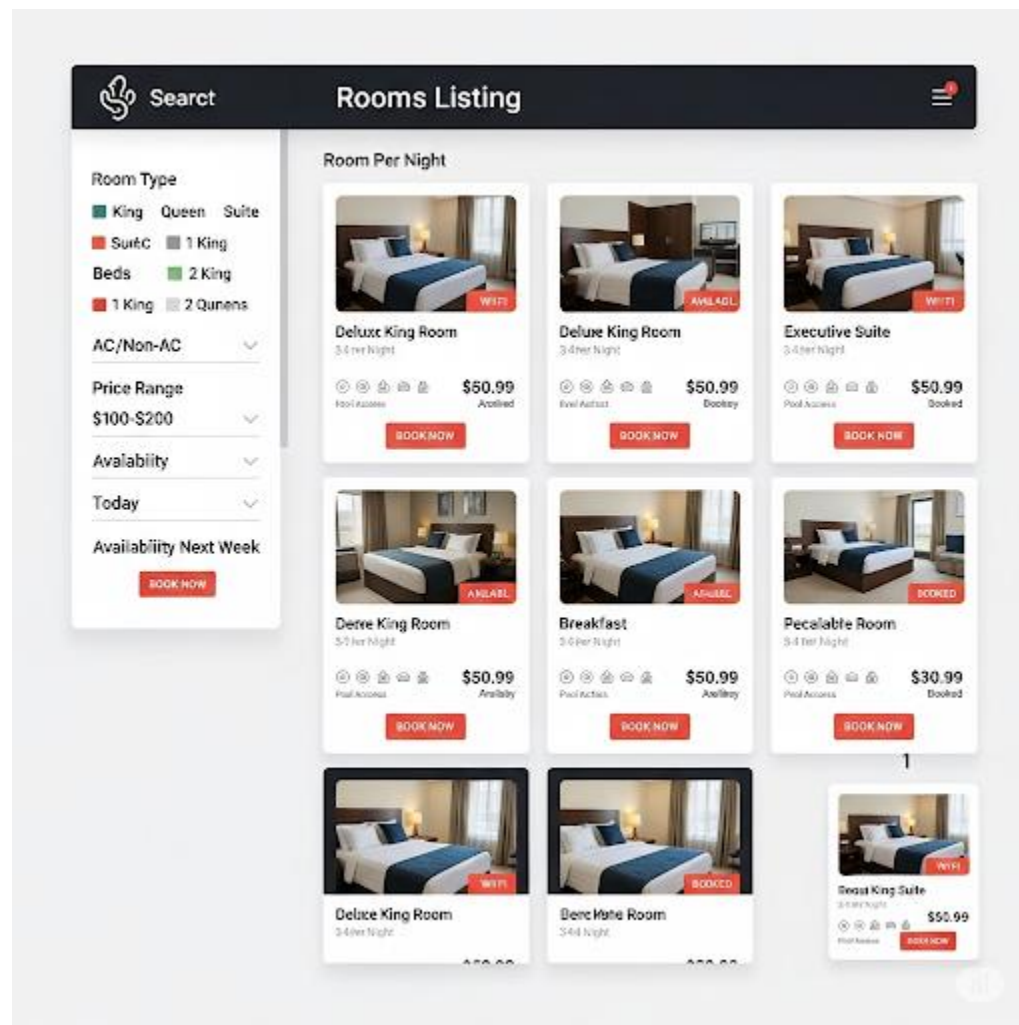
1.3.1. Homepage

- **Structure:** Hotel branding/logo, Navigation bar (Home, Rooms, Gallery, Contact, Login/Signup), Hero banner (carousel of hotel images or promo).
- **Features:** Introduction to hotel/services, Embedded guest reviews, Real-time room availability summary (from HMS backend).
- **Call-to-action:** "Book Now," "See Rooms," "Contact Us."
- [Image 15] - Website Homepage (from ui_ux (3).pdf, Page 18)
- [Image 16] - Website Homepage (from Untitled document (8).pdf, Page 1)



1.3.2. Rooms Listing Page

- **Structure:** Filter Panel (Room Type, AC/Non-AC, Beds, Price Range, Availability), Card Grid (Room Name/Type, Price per night, Photo gallery preview, Amenities icons, Availability badge, "Book Now" Button).
- **Backend Link:** Room status pulled from HMS room module; photos managed from desktop backend or hotel website management page.
- [Image 17] - Website Rooms Listing Page (from ui_ux (3).pdf, Page 19)
- [Image 18] - Website Rooms Listing Page (from Untitled document (8).pdf, Page 2)



1.3.3. Room Details Page

- **Structure:** Full-sized images (carousel or zoomable gallery), Room details (Description, Bed configuration, Max occupancy, Services included), Real-time availability calendar.
- **Call-to-action:** "Proceed to Booking" Button.
- [Image 19] - Website Room Details Page (from ui_ux (3).pdf, Page 20)
- [Image 20] - Website Room Details Page (from Untitled document (8).pdf, Page 3)

≡ Room Details



Bed Description

High lora the descriptions condridutions the amacle meer room the cervicnerathe Wish, bul cepanded afuo room and room sheghe preuries.

Bed Configuration

1 2 beds (Type)

Maximum Occupation

Maxt Occupation Allowed

Included Services

Included Services

- ✓ Complimentary Wi-Fi
- Complimentary Wi-Fi
- [Clicker Date @ Travel or Bookings](#)

2,33 Beds

Crotils

Breakfast

In-room Dining

Real-Time Availability

Sun	Mo	Tu	W	Th
6	6	7	0	2
14	15	16	17	18
21	22	23	25	26
27	29	30		

Proceed to Booking



1.3.4. Booking & Payment Page

- **Booking Form:** Guest details (Name, Phone, Email, Nationality, ID Upload), Check-in/Check-out date pickers, Room selector, Optional add-ons (Breakfast, Airport pickup, Early check-in), Total price auto-calculated.
- **Payment Section:** Payment modes (UPI, Credit/Debit Card, Netbanking), QR Code Generation (linked to HMS billing module), Email confirmation with invoice attached.
- **Integration:** Creates booking record in desktop HMS; triggers update in Room status + Upcoming Bookings table.
- [Image 21] - Website Booking & Payment Page (from ui_ux (3).pdf, Page 21)

Booking & Payment Sudags [Shiapor](#)

Guest Details

Name

Email

Phone

Nationality

Guest Details

Email

Check-in

Check Out

Room Select

Room	Room	Room	Room	Room	Room	Wegall	Total Price
Spom	Spom	Spom	Spom	Spom	Spom	Spom	\$22000

Optional Add Ons

Breakfast ☒ Airport Transfer ☐

Airport Transfer ☐ Spa ☐ Spaservices ☐

Payment

☒ UPI ☐ Credit/Debit Card ☐ Netbanking ☐ Netbanking ☐ QR Code

Email Confirmation

☐ Receive instant booking confirmation is email.

☐ Payment confirmation

☐ Payment your cofchations, 1\$2230 the opt in

☐ Hotel information

[Sal ouder prucer](#)

[Checkings](#)

1.3.5. Complaint / Service Request Page

- **Structure:** Guest fields (Name, Booking ID, Email/Phone), Complaint/Request Type (Room Cleanliness, Extra Towel/Pillow, Maintenance, Other), Textarea + optional photo upload.
- **Backend:** Tickets appear in HMS Guest Experience module or Service page.
- **Action:** Submit button creates a ticket forwarded to Housekeeping or Manager.
- [Image 23] - Website Complaint/Service Request Page (from ui_ux (3).pdf, Page 22)
- [Image 24] - Website Complaint/Service Request Page (from Untitled document (8).pdf, Page 5)

The screenshot shows a web form titled "Hoter Complaint// Service Request" (note the typo "Hoter"). At the top left is a logo with a house icon and the word "HOTEL". At the top right is a purple button labeled "Suest Now" (note the typo "Suest"). The form contains the following fields and elements:

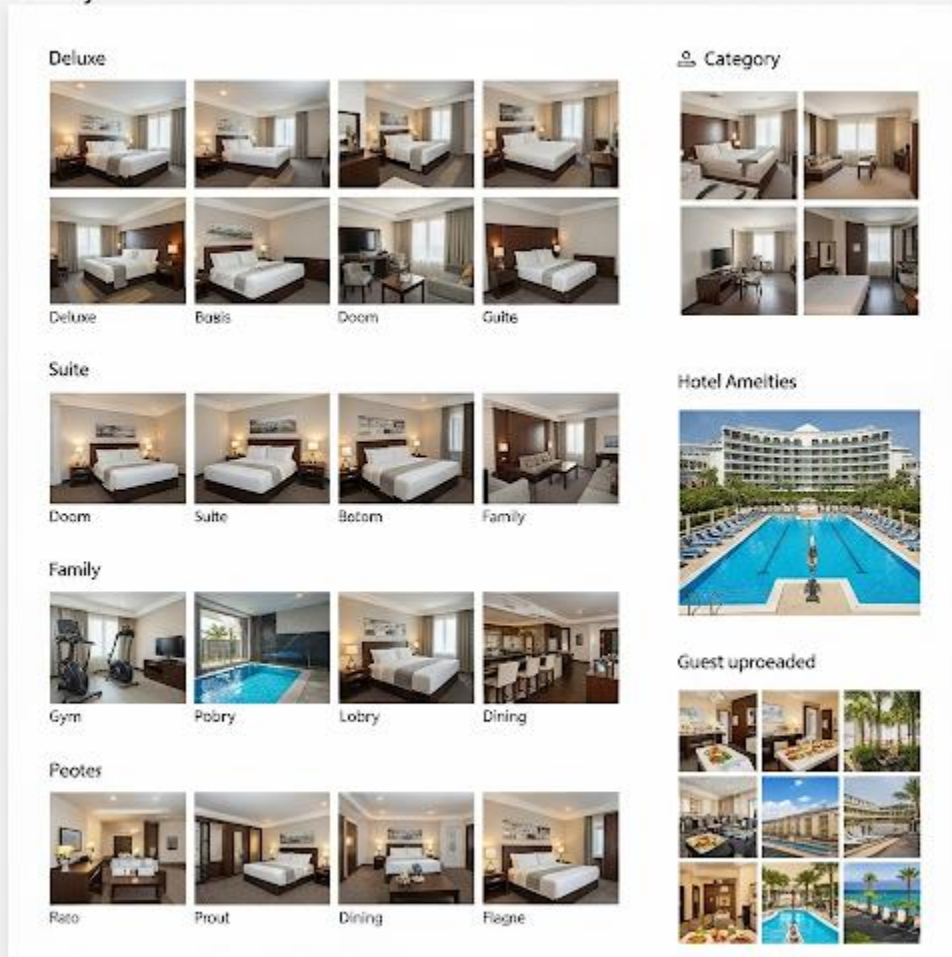
- Guest information:** A section header above the first set of fields.
- Full Name:** A text input field with a dropdown arrow on the right.
- Email (ogeniema):** A text input field (note the typo "ogeniema").
- Booking ID:** A text input field.
- Phone number:** A text input field.
- Complaint/Request:** A dropdown menu with a search icon. The menu is open, showing three options: "Room cleainieiss" (note the typo "cleainieiss"), "Extro towels or pillows" (note the typo "Extro"), and "Maintenance Issues".
- Specific needs:** A section header above the "Other" field.
- Other:** A text input field.
- Smarr photo upload:** A text input field (note the typo "Smarr").
- Optional photo upload:** A purple button at the bottom of the form.

Below the "Smarr photo upload" field, there is a paragraph of placeholder text: "Your text Secciant tor dece With Text Complamt yous dectiring wewe for hques your the your meestqur ichoinest to, stants sues unq your hely met upload" (note the numerous typos).

1.3.6. Gallery Page

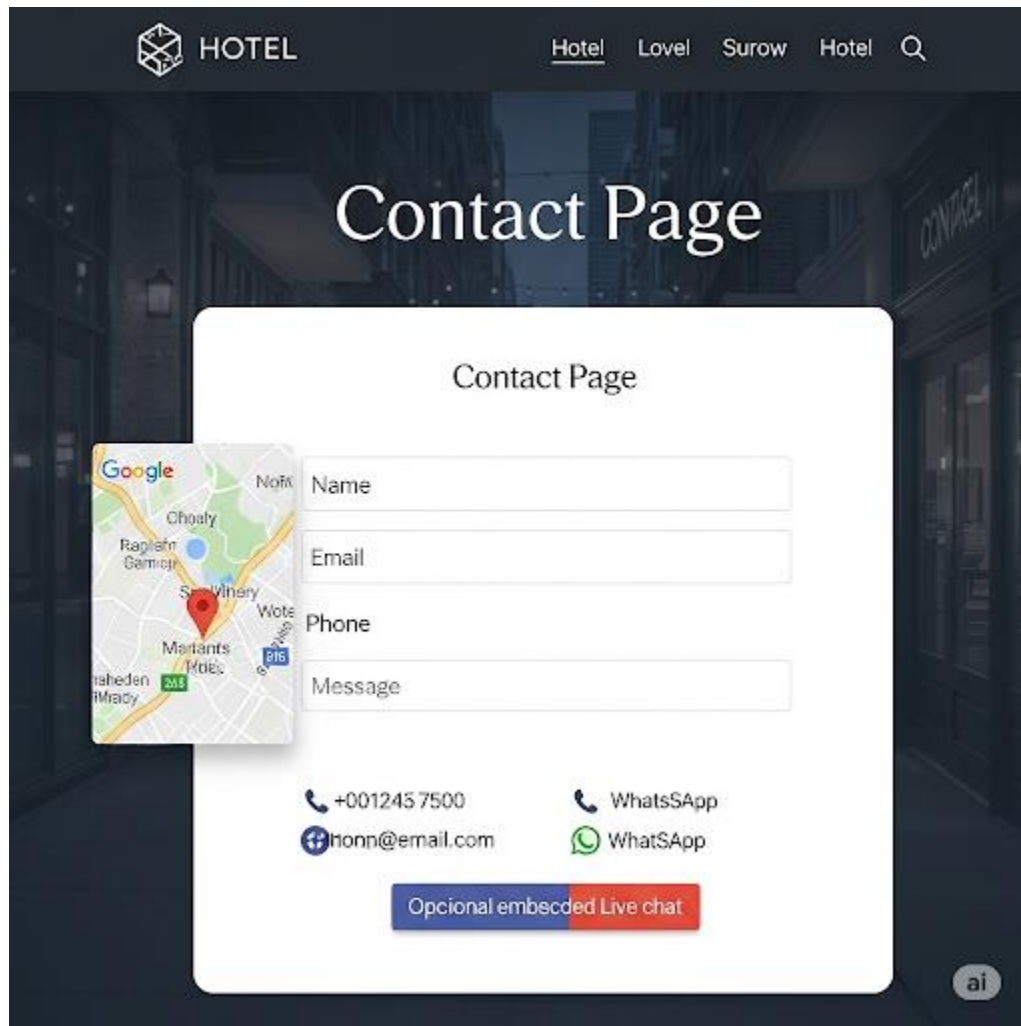
- **Structure:** Room photos grouped by category (Deluxe, Suite, etc.), Hotel amenities (Gym, Pool, Lobby, Dining, etc.), Optional guest-uploaded images (with moderation).
- [Image 25] - Website Gallery Page (from ui_ux (3).pdf, Page 23)
- [Image 26] - Website Gallery Page (from Untitled document (8).pdf, Page 6)

Gallery



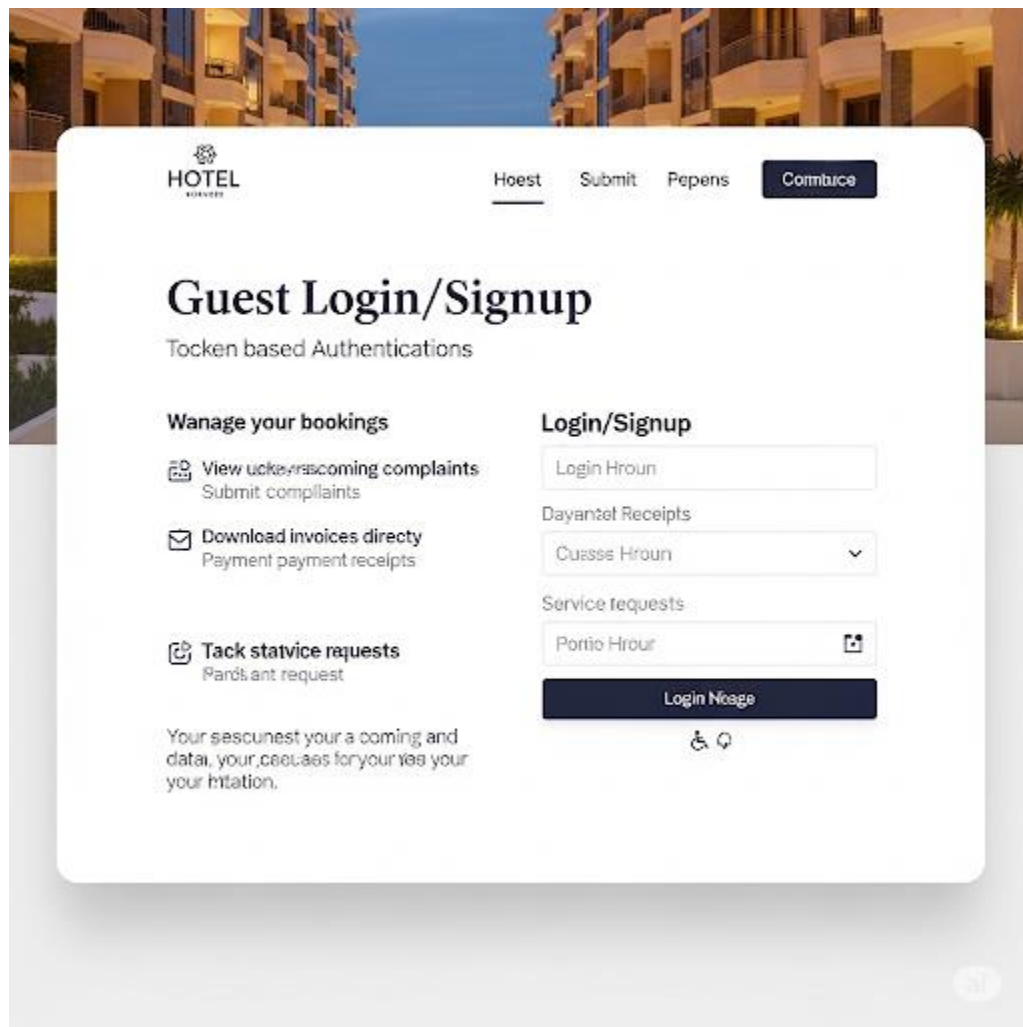
1.3.7. Contact Page

- **Structure:** Google Maps integration (hotel location), Contact Form (Name, Email, Phone, Message), Phone/Email/WhatsApp links, Embedded live chat (optional).
- [Image 27] - Website Contact Page (from ui_ux (3).pdf, Page 24)
- [Image 28] - Website Contact Page (from Untitled document (8).pdf, Page 7)



1.3.8. Guest Login/Signup Page

- **Purpose:** Guests can manage bookings, complaints, and payment receipts.
- **Features:** View upcoming bookings, Download invoice, Track service requests.
- **Backend:** Linked to Booking ID + phone/email; Simple token-based authentication.
- [Image 29] - Website Guest Login/Signup Page (from ui_ux (3).pdf, Page 25)
- [Image 30] - Website Guest Login/Signup Page (from Untitled document (8).pdf, Page 8)



1.4. Website-HMS Integration Summary

The table below summarizes how different pages of the public website connect with the Hotel Management System modules:

Website Page	Connected HMS Module(s)
Room Listings & Details	Rooms Module + Photo Storage
Booking + Payment	Booking Module + Checkout & Billing Module
Complaint/Service Request	Guest Experience + Housekeeping Module
Gallery	Hotel Website Management (Photo Storage)
Contact Page	External Only (Optional Logs)
Guest Login	Booking + Billing Record Viewer

2. Backend: System Logic and Data Management

The Backend of the HMS is the core logical layer responsible for processing requests, managing business rules, and interacting with the database.

2.1. Project Overview

This document provides a comprehensive overview of the backend architecture of the Hotel Management System (HMS), designed to support a JSP-based website, a JavaFX desktop application, and a React Native mobile app. The system is modular, scalable, cost-efficient, and built using modern technologies to ensure high availability, performance, and maintainability.

2.2. Technologies Used (Backend Specific)

- **Backend Framework:** Spring Boot (Java 17+)
- **Security:** Spring Security + JWT (for API clients), Session-based (for JSP website)
- **Hosting:** Railway.app or Render (for backend APIs)
- **Monitoring:** Prometheus (optional integrations)
- **Caching:** Optional Redis caching layer



Hotel Management System

HMS



Spring Security



Spring Security

Spring Security
JWT

Microservice

Spring Security
JWT

Microservice

Spring Security
JWT

Database Service

Spring Security
JWT

Guest Services

Microservice: Authentication, Authorization, User Management, etc.
Database Service: JWT (for API client) + Redis (for frequently accessed data)
Guest Services: Booking, Room Management, etc.



Redis



Hosting



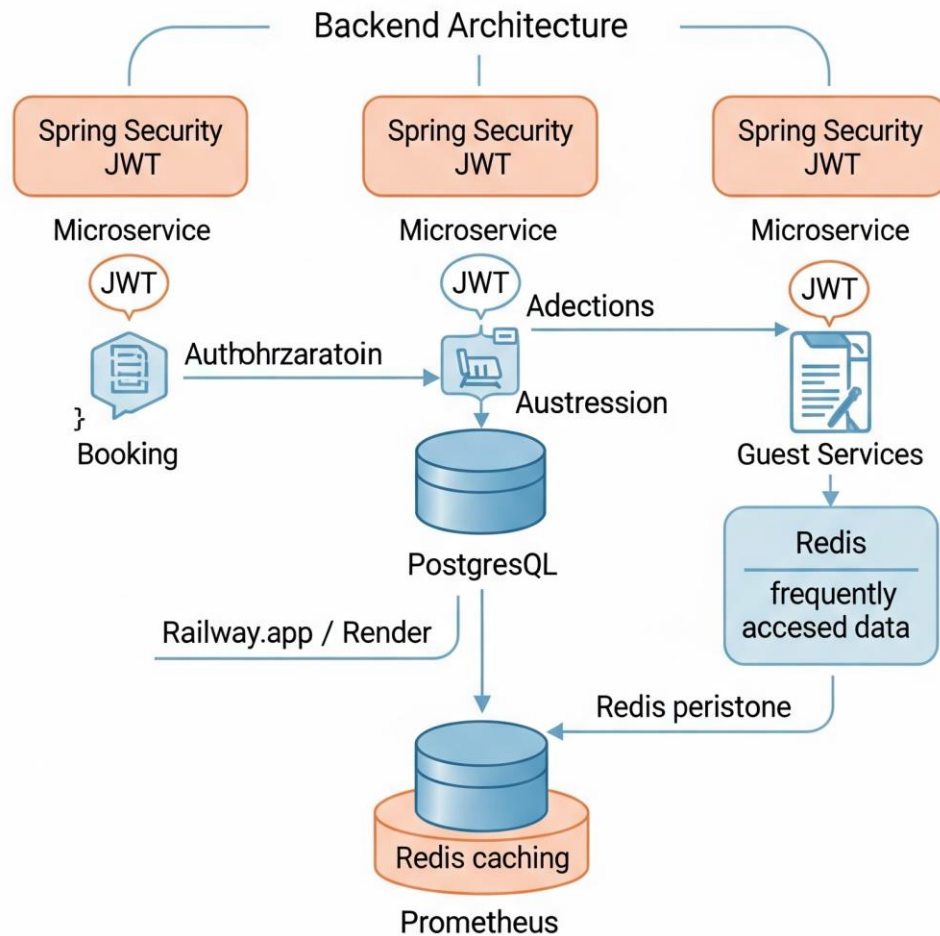
Redis
frequently
accessed data

Redis



Railway.app or Render PaaS





2.3. Backend Architecture and Modules

The HMS backend comprises the following key modules:

- **Auth Module:**
 - Handles authentication and authorization.
 - Supports JWT for API clients (mobile, desktop) and session for JSP website.
 - Roles: OWNER, MANAGER, RECEPTIONIST, STAFF, GUEST.
 - Enforces access control via Spring Security annotations.
- **Booking Module:**

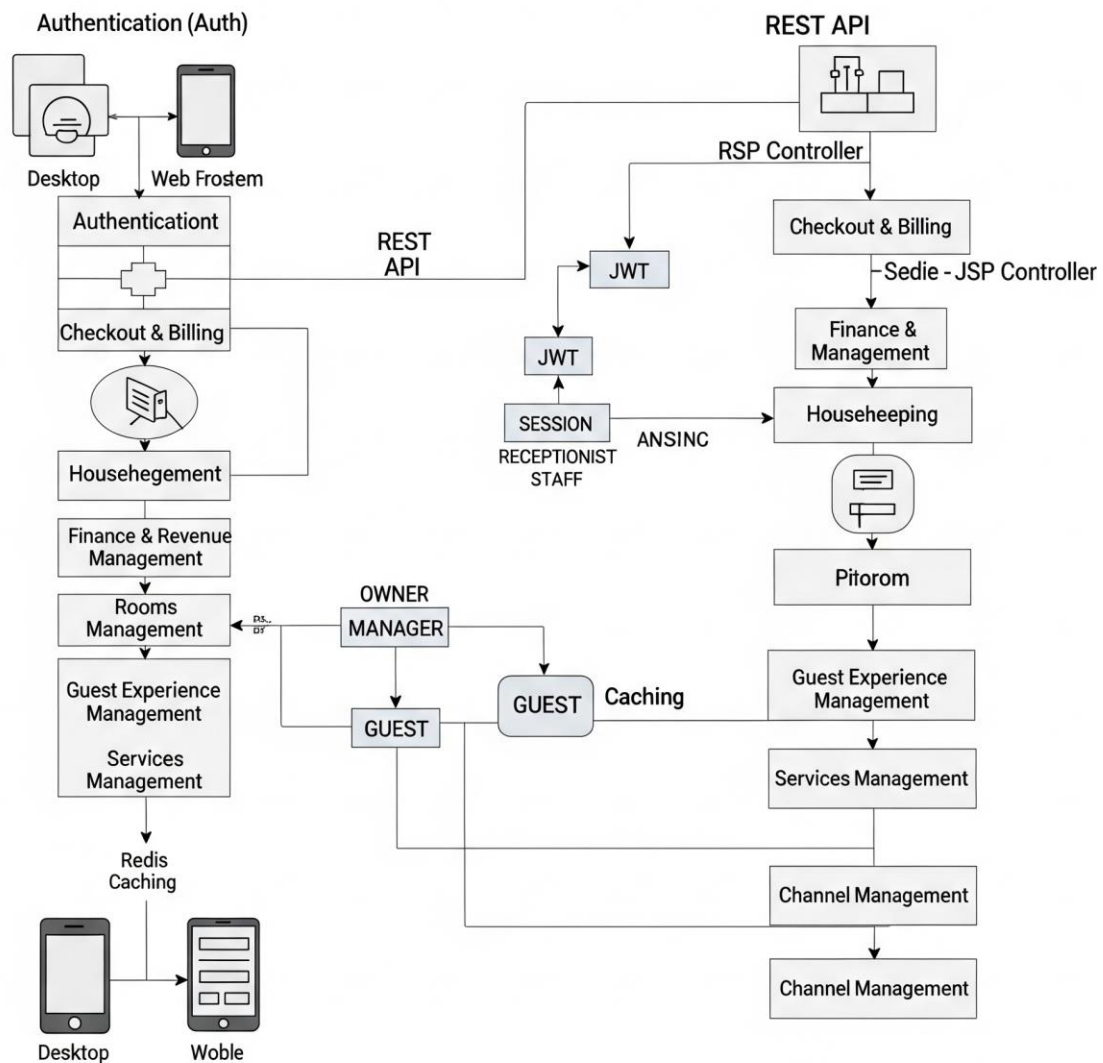
- Manages room availability, creates bookings, and calculates total cost.
- Auto-locks rooms on booking to avoid conflicts.
- Validates dates, occupancy, and guest information.
- **Checkout & Billing Module:**
 - Finalizes the stay, calculates remaining balance, applies charges, and generates GST-compliant invoices.
 - Integrates with Services & Inventory modules.
 - Generates QR codes for payment.
- **Rooms Module:**
 - Provides CRUD (Create, Read, Update, Delete) operations for room management.
 - Tracks room status: Occupied, Empty, Cleaning, Maintenance.
 - Linked to Booking, Checkout, and Housekeeping modules.
- **Housekeeping Module:**
 - Assigns and updates cleaning/maintenance tasks.
 - Tasks have statuses: Pending, In Progress, Done.
 - Staff can only see assigned tasks; Managers see all tasks.
- **Inventory Module:**
 - Tracks stock of room supplies.
 - Allows assigning items to rooms.
 - Alerts on reorder levels.
- **Finance & Revenue Module:**
 - Logs expenses, salaries, and calculates net profit.
 - Generates monthly revenue and cost reports.
 - Visualized in the owner dashboard.
- **Guest Experience Module:**
 - Handles guest feedback, complaints, and service requests.
 - Complaints can be routed to specific departments (e.g., Housekeeping).
 - Tracks the status of resolution for complaints and requests.
- **Camera Feed Module:**
 - Stores and retrieves RTSP or streaming URLs per room.
 - Accessible to owners/managers only.
 - Used by the mobile app for live hotel monitoring.
- **Services Module:**
 - Manages paid services like food, laundry, and car rental.
 - Linked to bookings and invoices.
 - Auto-added to the checkout bill if consumed.
- **Channel Management Module (Optional):**

- Enables OTA (Online Travel Agency) synchronization (e.g., Axis Rooms).

2.4. Backend Data Flow

All frontends (desktop, mobile, web) communicate with the unified Spring Boot backend via R

Hotel Management System

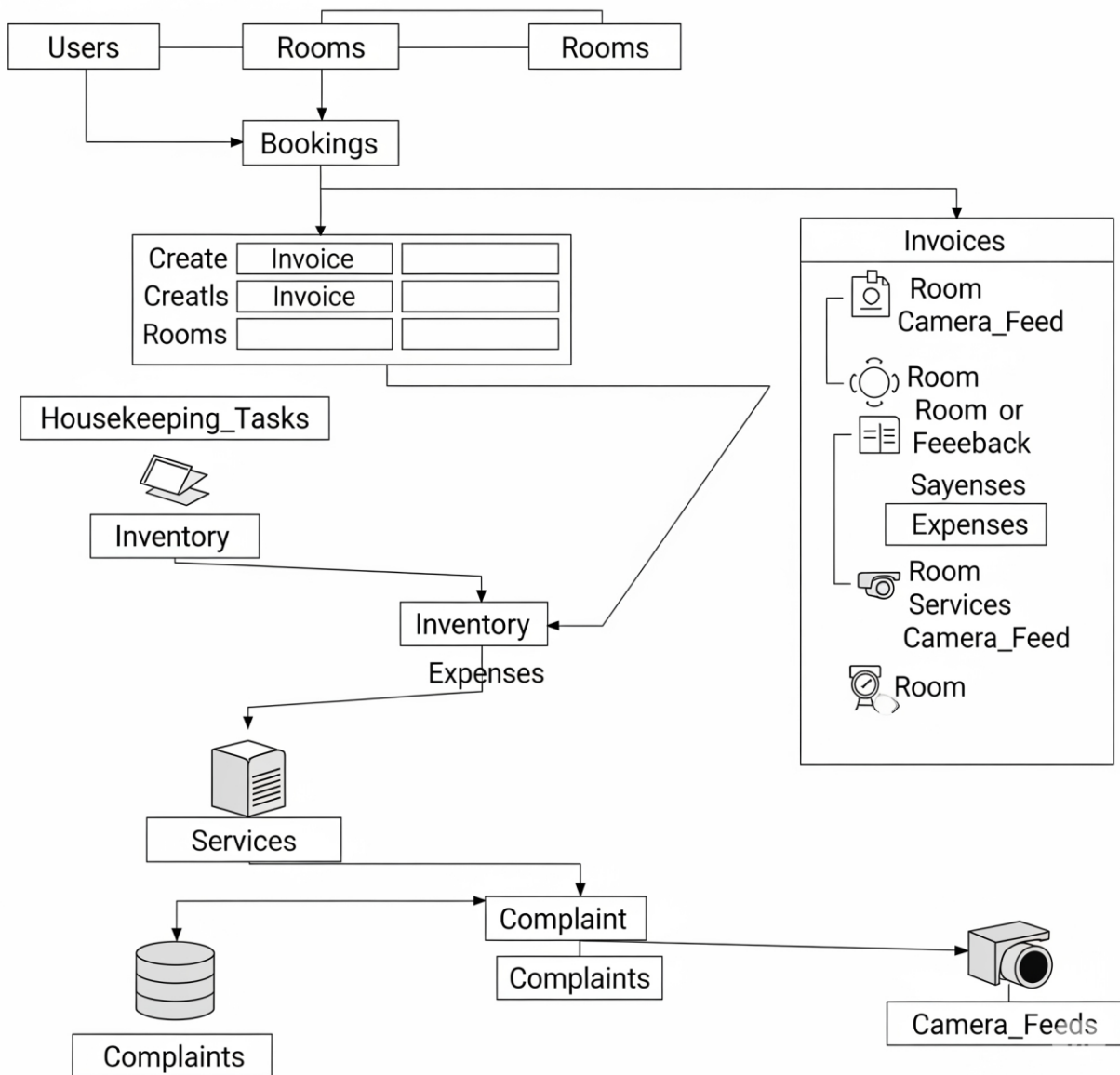


EST APIs or JSP controllers.

- Authentication is role-based using Spring Security and JWT for mobile/desktop applications, and session-based for the web application.
- An optional Redis caching layer is available for performance optimization but is decoupled and can be toggled off.

3. Database: Data Storage and Persistence

The Database component of the HMS is responsible for storing, retrieving, and managing a Entity-Relationship ERD



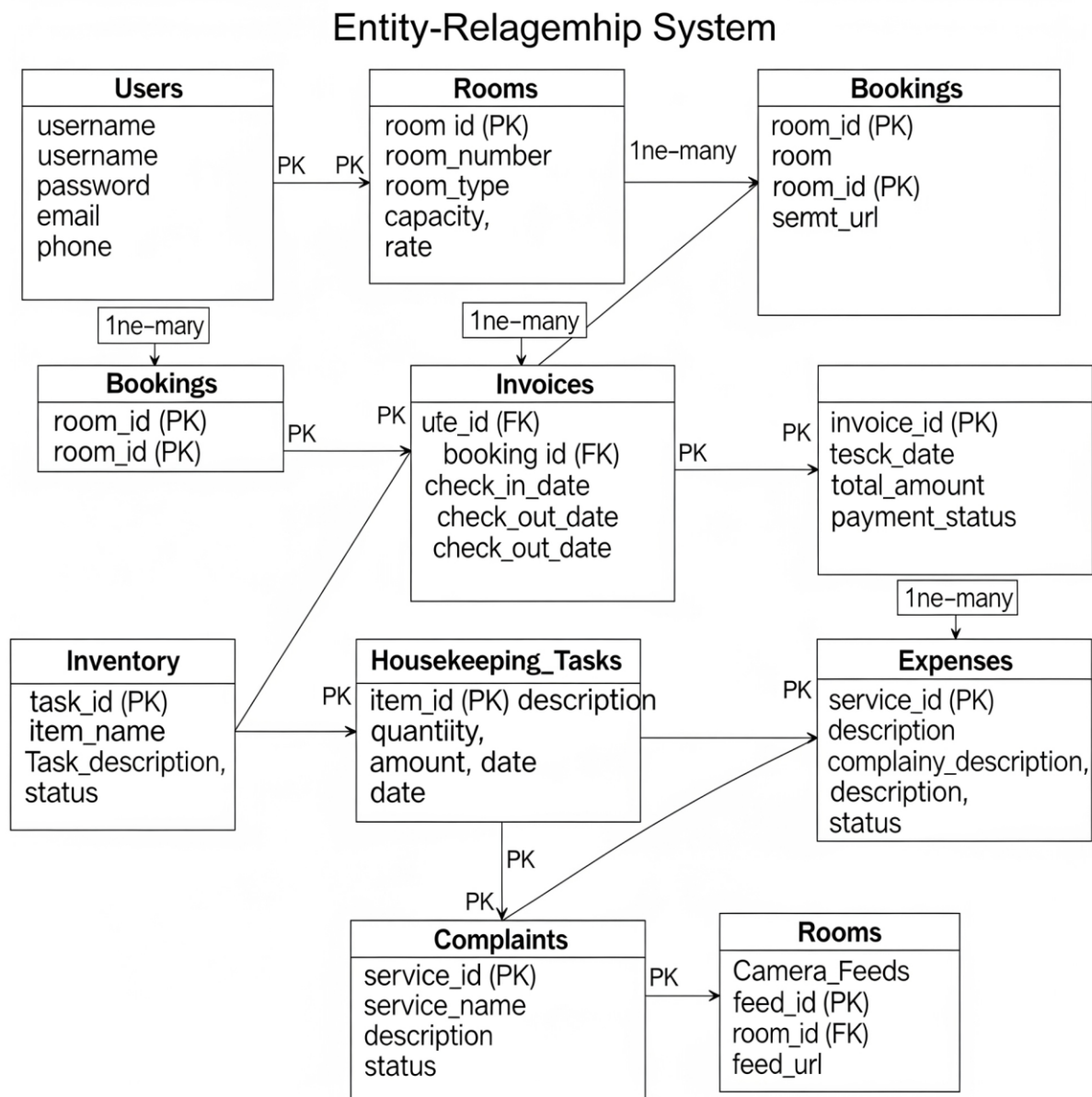
ll critical application data persistently.

3.1. Database Technologies Used

- **Primary Database:** PostgreSQL
- **File Storage:** Supabase or AWS S3 (for file storage like ID scans, guest photos)

3.2. Database Schema Overview

The PostgreSQL database schema includes multiple normalized tables designed for scalability and maintainability.

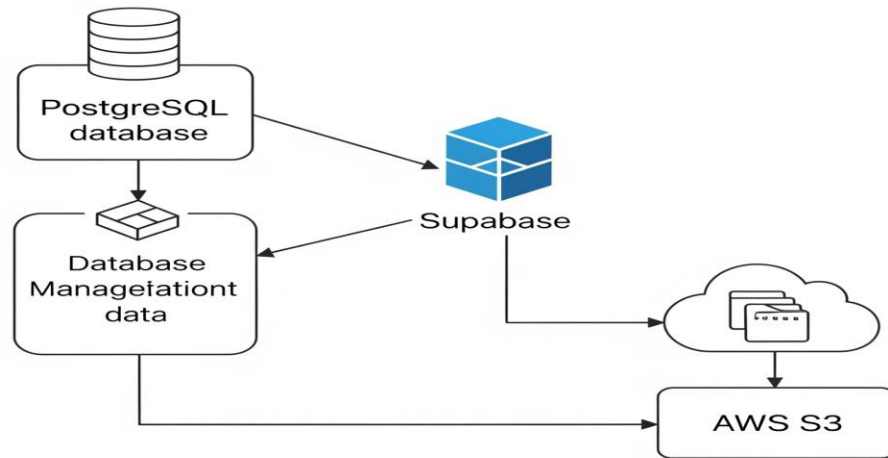


3.2.1. Core Tables

- users:(id, name, email, password, role, hotel_id)
- rooms:(id, number, type, floor, status, beds, price, ac, hotel_id)
- bookings:(id, guest_id, room_id, checkin, checkout, status, total_amount)
- invoices:(id, booking_id, issued_at, total, payment_status)
- housekeeping_tasks:(id, room_id, assigned_to, task_type, status, created_at)
- inventory:(id, item_name, quantity, reorder_level, room_id)
- expenses:(id, title, amount, department, created_at)
- services:(id, booking_id, type, amount, status, created_at)
- complaints:(id, guest_name, room_id, type, description, status)
- camera_feeds:(id, room_id, stream_url)

3.3. Hosting & Infrastructure (Database Specific)

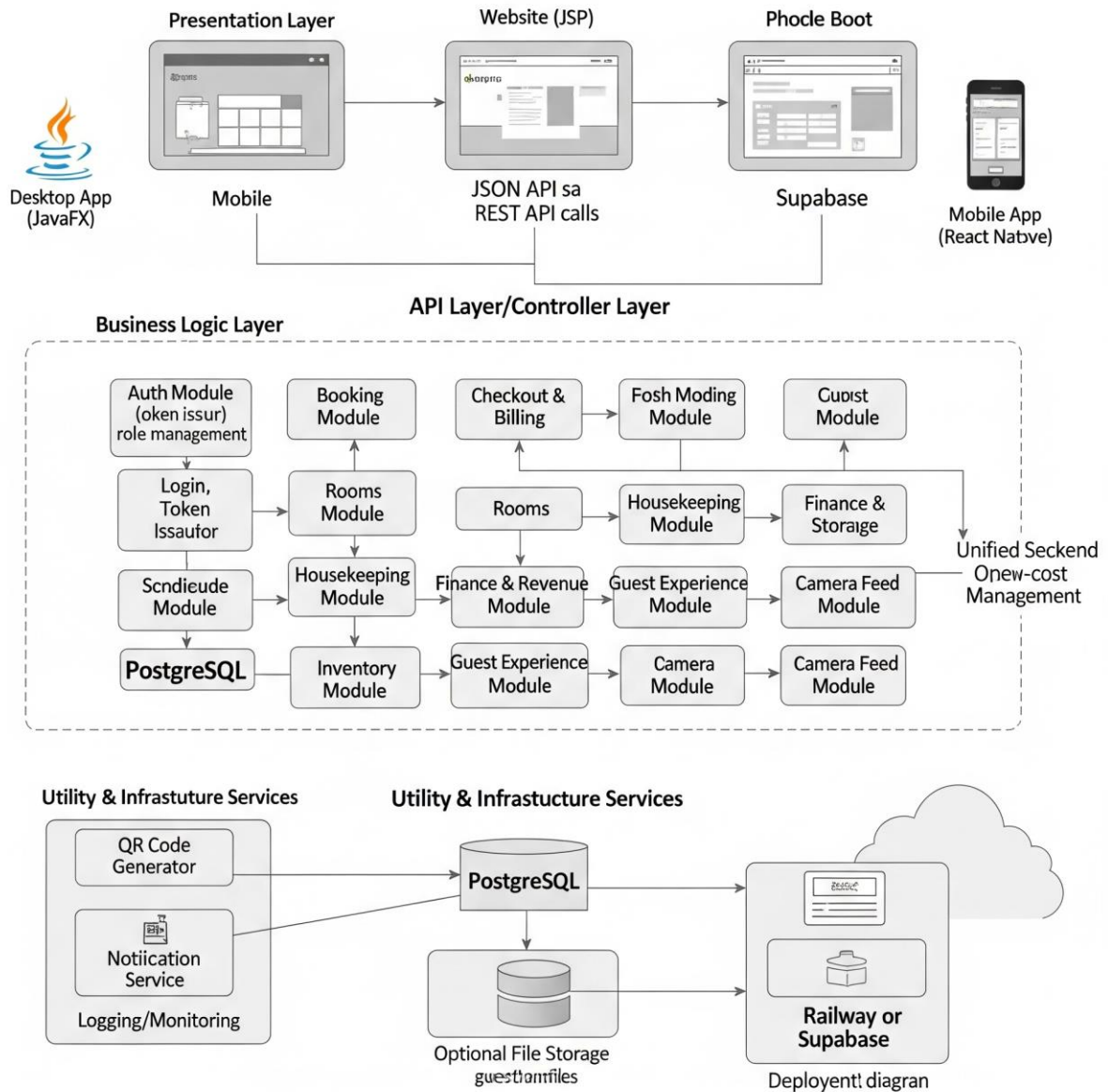
- **PostgreSQL Database:** Provided by Supabase (available in free or paid tiers).
- PostgreSQL stores all critical relational data.
- Supabase or AWS S3 handles optional file storage (e.g., ID scans or guest photos).



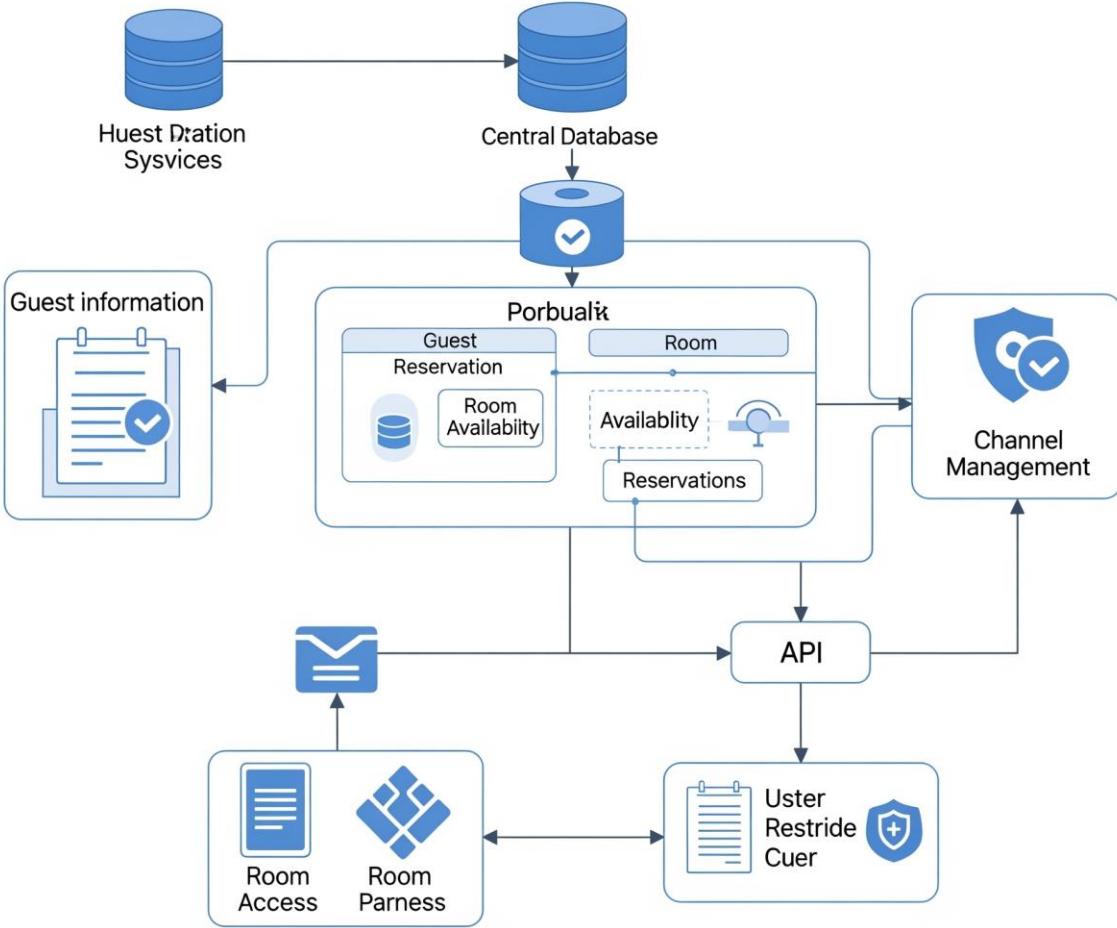
3.4. System Architecture Diagram

Below is a conceptual diagram illustrating the backend architecture and its integration with various frontend applications, specifically highlighting the database's role.

Hotel Management SyMS



Backend Management system



Hotel Management System

 Spring Boot

